

Inclusive, Exclusive and Hierarchical Atlas of NFATc1⁺/PDGFR-α⁺ Cells in Dental and Periodontal Mesenchyme

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Utilizing transgenic lineage tracing techniques and tissue clearing-based advanced imaging and three-dimensional slices reconstruction, the authors comprehensively mapped the distribution atlas of NFATc1⁺ and PDGFR-α⁺ cells in dental and periodontal mesenchyme and tracked their in vivo fate trajectories. This **important** work extends our understanding of NFATc1⁺ and PDGFR-α⁺ cells in dental and periodontal mesenchyme homeostasis, and should provide impact on clinical application and investigation. The strength of this work is **compelling** in employing CRISPR/Cas9-mediated gene editing to generate two dual recombination systems, and mapped gNFATc1⁺ and PDGFR-α⁺ cells residing in dental and periodontal mesenchyme, their capacity for progeny cell generation, and their inclusive, exclusive and hierarchical relations in homeostasis, generating a spatiotemporal atlas of these skeletal stem cell population.

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Abstract

Platelet-derived growth factor receptor alpha (PDGFR-α) activity is crucial in the process of dental and periodontal mesenchyme regeneration facilitated by autologous platelet concentrates (APCs), such as platelet-rich fibrin (PRF), platelet-rich plasma (PRP) and concentrated growth factors (CGF), as well as by recombinant PDGF drugs. However, it is largely unclear about the physiological patterns and cellular fate determinations of PDGFR-α⁺ cells in the homeostasis maintaining of adult dental and periodontal mesenchyme. We previously identified NFATc1 expressing PDGFR-α⁺ cells as a subtype of skeletal stem cells (SSCs) in limb bone, but their roles in dental and periodontal remain unexplored. To this end, in the present study we investigated the spatiotemporal atlas of NFATc1⁺ and PDGFR-α⁺ cells residing in dental and periodontal mesenchyme, their capacity for progeny cell generation, and their inclusive, exclusive and hierarchical relations in homeostasis. We utilized CRISPR/Cas9-mediated gene editing to generate two dual recombination systems, which were

Cre/loxP-Dre/Rox combined intersectional and exclusive reporters respectively, to concurrently demonstrate the inclusive, exclusive, and hierarchical distributions of NFATc1⁺ and PDGFR- α ⁺ cells and their lineage commitment. By employing the state-of-the-art transgenic lineage tracing techniques in cooperating with tissue clearing-based advanced imaging and three-dimensional slices reconstruction, we systematically mapped the distribution atlas of NFATc1⁺ and PDGFR- α ⁺ cells in dental and periodontal mesenchyme and tracked their *in vivo* fate trajectories. Our findings extend current understanding of NFATc1⁺ and PDGFR- α ⁺ cells in dental and periodontal mesenchyme homeostasis, and furthermore enhance our comprehension of their sustained therapeutic impact for future clinical investigations.

Introduction

Platelet-derived growth factor (PDGF), first identified in platelets, intricately involved in the process of tissue regeneration facilitated by autologous platelet concentrates (APCs)^{1, 2}. The effects of PDGFs are mediated by two receptor tyrosine kinases, namely PDGFR α and PDGFR β . The PDGFR α homodimer predominantly binds to the majority of PDGF isoforms (PDGF-AA, PDGF-AB, PDGF-BB, and PDGF-CC)³. In recent decades, APCs have gained widespread utilization in the field of tissue regenerative medicine, with notable generations including platelet-rich plasma (PRP), platelet-rich fibrin (PRF), and concentrated growth factors (CGF). Evidence-based medicine studies have demonstrated that these three APC techniques have been utilized in the field of periodontology, contributing to periodontal regenerative therapy, albeit with varying efficacy⁴. Additionally, certain investigations suggest that PDGF-BB also exerted a role in promoting regeneration of both periodontal tissues and dental pulp^{5, 6}. Teeth, being highly mineralized organs, enclose a mineralized chamber housing the dental pulp mesenchyme, while the surrounding tissue comprises periodontal soft tissue—periodontal ligament (PDL)⁷. Simply, teeth can be regarded as hard tissues enveloped by mesenchyme. While the dental pulp and periodontium fulfill distinct physiological roles, they also maintain interconnectedness to collectively sustain the physiological function of the tooth. Consequently, when pathological changes occur independently or concurrently in the dental pulp and periodontal tissues, they may mutually influence each other⁸. Therefore, it is imperative to elucidate the expression and biological mechanism of PDGFR- α ⁺ cells in PDL and dental pulp tissues in order to comprehensively comprehend the physiological and therapeutic effects mentioned above.

PDGFR- α serves as a pivotal signaling molecule involved in mediating tooth development, morphogenesis, and physiological responses^{9–11}. Its potential role in periodontal formation has also been suggested¹². Within skeletal tissues, PDGFR- α has been identified as a heterogeneous population, with not all PDGFR- α ⁺ cells exhibiting mesenchymal stem cell characteristics^{13–16}. Certain studies suggest that subpopulations of PDGFR- α ⁺ cells are considered “true” stem cells. For instance, co-labeling of Nestin and PDGFR- α has been observed in bone marrow stem cells (BMSCs) derived from the neural crest¹⁷, and the co-expression of PDGFR- α and Sca-1 in the non-hematopoietic compartment has identified a distinct cell population within the adult murine bone marrow¹⁸. Our prior investigations have also revealed a population of NFATc1 and PDGFR- α double-positive skeletal stem cells (SSCs) within long bones¹⁹. In long bones, NFATc1⁺ cells are considered SSCs¹⁹. Furthermore, previous RNA-sequencing analyses have reported the expression of NFATc1 in jawbones and periodontal tissues²⁰. However, this evidence was limited to RNA-level sequencing, lacking data support *in vivo*.

The conventional single-enzyme cutting system, represented by the *Cre-loxP* system primarily focuses on individual cells²¹. The introduction of multiple DNA recombinases has revolutionized lineage tracing by enabling a layered and intersectional analysis that was previously unattainable^{22, 23}. Notably, the utilization of multiple recombinase-mediated recombination

includes systems such as Intersectional reports (IR) and exclusive reporters (ER), which facilitate the visualization of two sets of orthogonal recombination or hierarchical relationships²³. In this study, considering significant tissue variations in dental pulp and PDL, traditional slicing methods may inadequately represent their expression patterns²⁴. To ensure accurate depiction of expression patterns in spatial dimensions, we employed two strategies: tissue transparency (TC) and three-dimensional (3D) slices reconstruction, guaranteeing comprehensive spatial imaging technology for investigating tissue structural architecture.

In summary, we utilized the advanced *Cre/loxP-Dre/Rox* combined intersectional and exclusive reporters—IR and ER systems along with advanced imaging methods to comprehensively examine the intricate relationships, including inclusive, exclusive and hierarchical between two distinct cell populations within pulp and periodontal mesenchyme for the first time. Furthermore, we demonstrated their capacity for progeny cell generation and hierarchical positioning to depict their *in vivo* fate determination trajectories. Additionally, we assessed the advantages and limitations of TC and 3D slices reconstruction techniques. These findings not only furnish essential tools and references for future investigations into MSC lineage tracing within dental and periodontal tissues but also enhance our comprehension of biological research and therapeutic applications in regenerative medicine.

Result

1. Establishment of multiple genetic recombination systems and advanced imaging technology

To understand the cell fate, behaviors, and their spatiotemporal correlation of PDGFR- α^+ and NFATc1 $^+$ cells in dental and periodontal mesenchyme, systematic genetic fate atlas were established by lineage tracing based on genetic DNA recombination technology. In conventional single recombinase-mediated genetic readouts, the *Cre-loxP* system is widely used for mammalian gene editing with Cre recombinase catalyzing recombination between *loxP* sites. This system, often activated by a ubiquitous promoter like Rosa26 and H11, removes a transcription stop cassette post-recombination to allow for reporter gene expression (Figure 1A). However, the drawback of the conventional approach is that it is impossible to observe the genetic lineage and spatial relationship of two specific cell type at the same time, also, targeting two gene promoters in one cell population could be more precise than relying on a single promoter as commonly employed in the conventional reporter system. Given that, diverse dual recombinase-mediated genetic labeling systems have been developed to enhance the specificity and the number of cell types being labeled simultaneously. *Cre-loxP*, *Flp-*frt**, *Dre-rox*, and *Nigri-nox* have been respectively used for designing dual system. Using reporter as an entry site can categorize these systems into three different types for multiple recombinase-mediated fate mapping studies, including intersectional reporters, exclusive reporters, and nested reporters (Figure 1A-b, c). In this work, the most advanced transgenic technology, dual recombinase-mediated genetic labeling systems, including intersectional reporters and exclusive reporters, were constructed to simultaneously show the inclusive, exclusive and hierarchical distribution of PDGFR- α^+ /NFATc1 $^+$ cells in dental and periodontal mesenchyme. Specifically, multichromatic crossover intersectional reporters (H11-LGRT) systems, which structured as *loxP-Stop-loxP-ZsGreen-rox-Stop-rox-tdTomato*, enabled the tracing of three cell types: *Cre+&Dre-*, *Cre-&Dre+*, and *Cre+&Dre+*, corresponding to the colors ZsGreen, tdTomato, and yellow, respectively. This technology guaranteed us to precisely understand the respective and inclusive distribution of Pdgfr- α^+ , NFATc1 $^+$ cells and their progeny cells. In addition, IR1 system is an exclusive reporter expressed as *CAG-loxP-rox-Stop-loxP-ZsGreen-Stop-rox-tdTomato* (Figure 1A-c). The first *Cre-loxP* recombination results in ZsGreen expression and the removal of a *rox* site, preventing subsequent *Dre-rox* recombination in the same cell. This means that once a recombination event occurs, the expression of the

corresponding reporter gene eliminates a recognition site for the other recombination system, which could provide generous information of the hierarchical distribution of PDGFR- α^+ and NFATc1 $^+$ cells in dental and periodontal mesenchyme.

Due to the peculiarity of dental/periodontal tissues, including pulp and periodontal of maxillary and mandibular first molars (small volume, complex structure and uneven spatial distribution of SSCs in mesenchyme), single-slice images based on traditional frozen section and confocal microscope imaging were insufficient to accurately presume the practical situation of the organization. Therefore, the advanced tissue deep clearance using SUMIC procedure and 3D reconstruction was exploited to observe the overall spatial distribution of PDGFR- α^+ and NFATc1 $^+$ cells in dental and periodontal mesenchyme of transgenic mice with multiple DNA recombinases-based genetic lineage tracing system (**Figure 1C-a**). Simultaneously, as an irreplaceable imaging technology, traditional sectioning combined confocal imaging and procedure was also estimated (**Figure 1C-b**). These two 3D-reconstruction and imaging technologies complement each other to jointly address the spatial positioning and hierarchical relationships of PDGFR- α^+ , NFATc1 $^+$, and PDGFR- α^+ NFATc1 $^+$ cells from multiple perspectives.

2. Exploration of NFATc1 $^+$ /PDGFR- α^+ cells and their progeny cells in H11-LGRT system

First, we established *Cre-loxP* and *Dre-rox* reporter lineage-tracking model of PDGFR- α and NFATc1 in adult mice to observe the spatial location of PDGFR- α^+ and NFATc1 $^+$ cells in pulp and periodontal of maxilla & mandible M1. The pulse timing point we set was tamoxifen treated twice at day 1 and day 3, and then the mice were sacrifice to obtain maxilla and mandible on day 5 (**Figure 2A, B**).

2.1 Dental and periodontal region of maxilla M1 and mandible M1

As one of the most advanced imaging technologies currently available, tissue clearing/imaging allows for direct observation of the spatial location and relationships of fluorescently labeled cells within the intact tissue. Therefore, according to the existing SUMIC tissue deep clearing (TC) methods^{25, 26}, we modified and improved a rapid and efficient procedure, which enable rapid single-cell resolution and quantitative panoptic 3D light-sheet imaging. This procedure not only rapidly clear maxillary hard tissue, but also directly penetrate the dentin to visualize the dental pulp. The total time of this procedure was only 2 days including tissue fixation and decalcification, this ultrafast clearing technique could minimize the impact on transgenic mice. As shown in Figure S1A-B, we recorded bright-field images of the maxilla before and after clearing, and our procedure achieved high transparency of the whole tissue. On this basis, whole-tissue imaging can be achieved, with the observation of different cell type distribution in spatial 3D structure. As such, dental pulp and periodontal ligament (PDL) region of M1 were representatively delineated for 3D reconstruction and analysis. **Figure 2C** and Video S1 showed the contoured maxilla M1, including pulp and periodontal ligament (PDL) with virtual dentin shell (white). The buccal view, coronal view, and radicular view of pulp and PDL showed that PDGFR- α^+ cells were equidistributional in the whole area of M1 pulp (including crown pulp and root pulp) and PDL area, almost constitute the whole dental and periodontal tissue, while NFATc1 $^+$ cells were almost exclusively distributed in crown pulp and scattered existence in PDL. Across the board, the distribution of NFATc1 $^+$ cells were significantly less than that of PDGFR- α^+ cells in both pulp and PDL area, indicated that the expansive range of PDGFR- α^+ cells which may contain NFATc1 $^+$ population. As mentioned above, studies have shown that with not all PDGFR- α^+ cells within long bones exhibiting mesenchymal stem cell characteristics, and only certain specific subpopulations of PDGFR- α^+ cells can be considered as “real” stem cells. In addition, our previous study has evidenced NFATc1 $^+$ PDGFR- α^+ cells as skeletal stem cells (SSCs) within long bones, played an indispensable role in bone homeostasis and regeneration. As such, these phenomena we observed

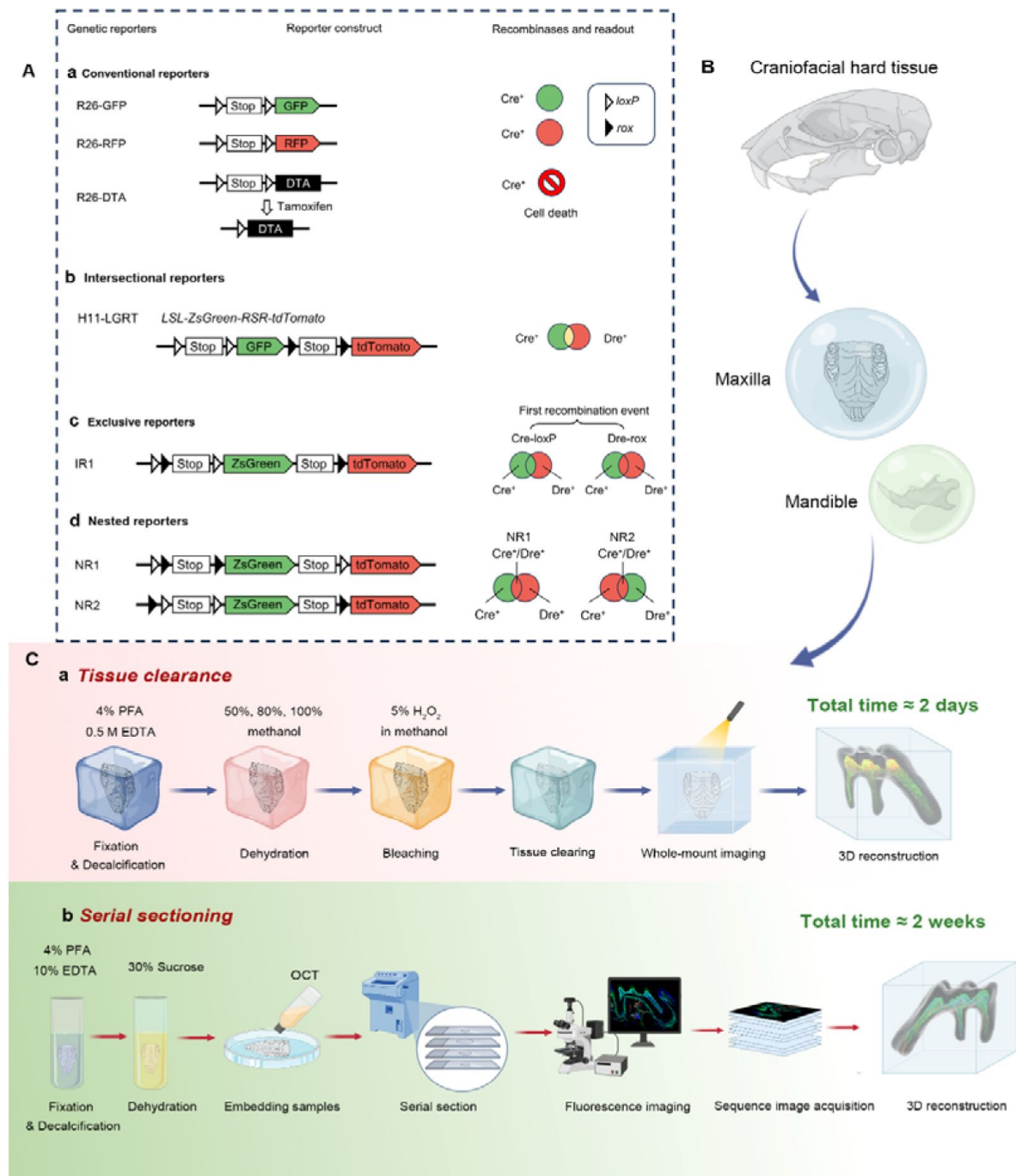


Figure 1.

(A) Genetic recombination systems. The initial column lists various reporter alleles, while the subsequent second and third columns present the corresponding reporter constructs and the recombinases utilized. The final column provides the outcomes associated with the respective reporter alleles. Traditional reporter mechanisms comprise both single-color and multicolor systems (not shown in the figure), created through a singular recombination event. Advanced dual-reporter configurations encompass intersectional reporters, exclusive reporters, and nested reporters. (B) The craniofacial hard tissue, including maxilla and mandible of transgenic mice with multiple DNA recombinases-based genetic lineage tracing system, were obtained for advanced imaging technologies. (C-a) The tissue deep clearing procedure based on SUMIC, which needed to go through 4% PFA fixation, 0.5M EDTA decalcification, different concentration of methanol dehydration, 5% H₂O₂ bleaching, ECI tissue clearing, and finally used the light-sheet system to directly image the whole tissue. (C-b) The traditional frozen section technology needs fixation, decalcification, dehydration and other steps, which takes about two weeks.

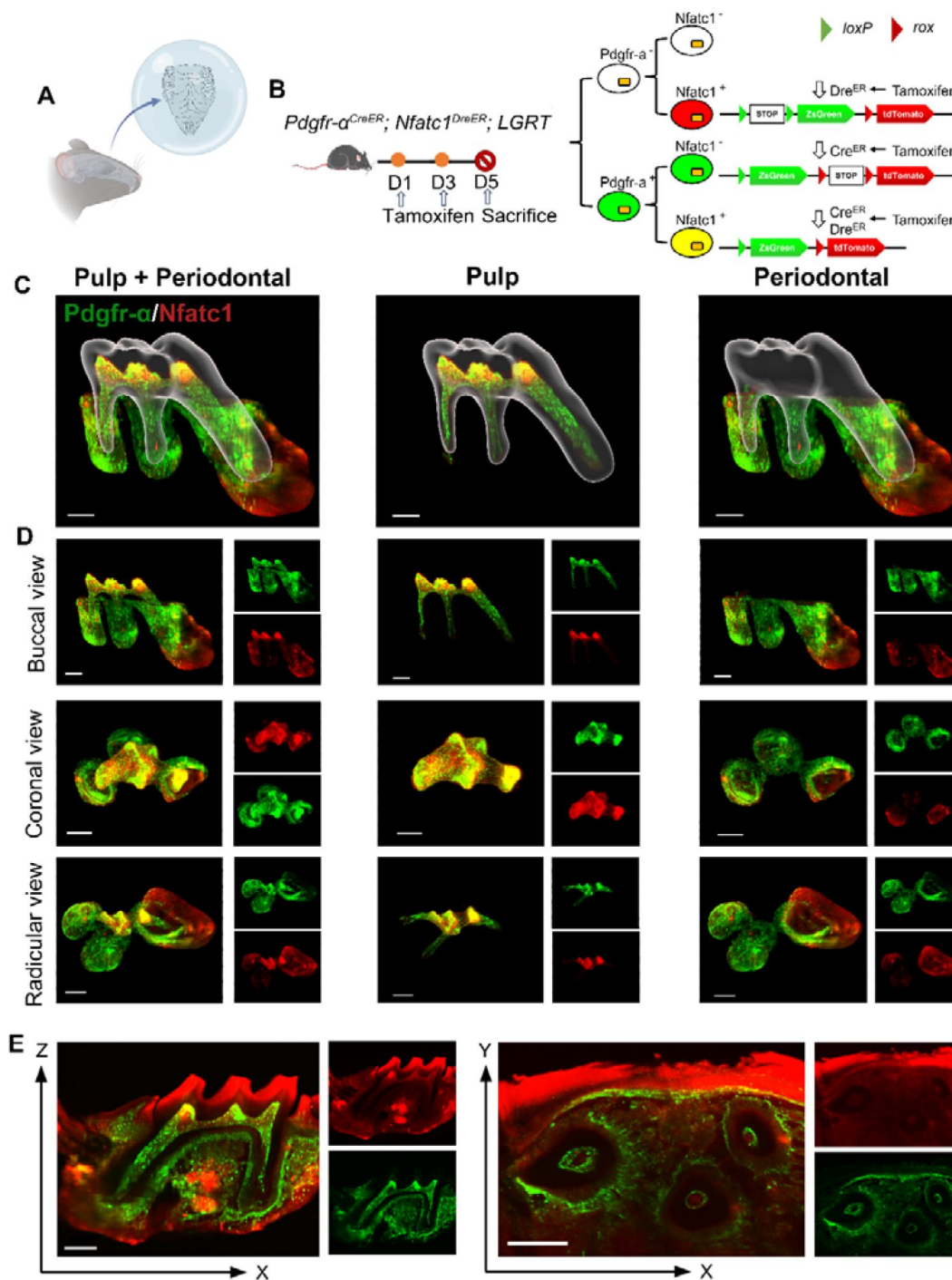


Figure 2.

The whole-mount and high-speed imaging of (A) mouse molar with a tiling light-sheet microscope. (B) Schematic illustration of lineage tracing in *Pdgfr-α^{CreER}; Nfatc1^{DreER}; LGRT* mice. The mice were administrated with tamoxifen at D1 and D3, and sacrificed at D5. (C) The contoured M1 of maxilla, including pulp and PDL with virtual dentin shell (white) in buccal view (scale bar = 300 μm). (D) Image stack was displayed in buccal view, coronal view, and radicular view of pulp and PDL, respectively (scale bar = 300 μm). (E) An optical slice was acquired on the X-Z (scale bar = 300 μm) and X-Y direction (scale bar = 400 μm) to display the pulp and PDL.

from the results of TC preliminarily indicate a similar situation in dental and periodontal tissue, and the biological role of NFATc1⁺PDGFR- α ⁺ cells in homeostasis and regeneration situation is our major focus in the future research.

In addition, from the section of XZ axis after 3D reconstruction, NFATc1⁺ cells were mainly distributed in the alveolar bone area, including the surface edge of trabecular bone (Figure S1C), which was consistent with RNA-sequencing results in the previous study²⁰. For the cross-sectional analysis of the 3D reconstructed data, the maximum cross-section of XZ axis which containing the whole pulp and PDL and the XY axis of the root pulp were selected (Figure 2E). However, the image is not ideal, which due to: 1) the editing efficiency DNA recombinase-mediated lineage-tracing system have limitations; 2) the less amount presence of NFATc1⁺ cells in the tissue guaranteed the weak signals; 3) the as-describe TC method may cause poor penetration of the td-tomato fluorescence signals.

Therefore, in order to obtain high resolution image to understand more detailed information, we still need to rely on the traditional serial section technology (Figure 3A). A total of 121 slices were collected in this maxilla of *Pdgfr- α ^{CreER}; Nfatc1^{DreER}; LGRT* mice sample, from the jaw root PDL just appeared, to the buccal root PDL completely disappeared (10 μ m/slice) (Figure S2). As shown in Figure 3B and Figure S3, although time consuming, the images obtained by confocal imaging can clearly obtain the distribution of PDGFR- α ⁺ and NFATc1⁺ cells from the single cell level. However, it is difficult to determine the presence of PDGFR- α ⁺&NFATc1⁺ cells spatiotemporally. So, we also used Imaris to 3D-reconstruct these 121 images. Figure 3C and Video S2 showed the contoured M1 of maxilla, including pulp and PDL with virtual dentin shell (white). Figure 3D was the buccal view, coronal view, and radicular view of pulp and PDL, respectively. Different from the whole imaging obtained by TC and light-sheet, the td-tomato signal reconstructed by traditional serial section-based confocal imaging method was very conspicuous, even in root pulp, which were hardly obtained by TC imaging (Figure S4). Same as TC-based reconstructed results, PDGFR- α ⁺ cells almost constituted the whole structure of pulp and PDL, with NFATc1⁺ cells as subpopulation. However, due to the stratification of lices, the sample integrity was poor, leading to discontinuities in the z-axis. (Figure S5), which may render some biases in the results. To enhance the comprehensive and accurate display of the reconstruction results and to mitigate the potential errors that may arise from relying on single reconstruction method, we employed an alternative 3D reconstruction method—DICOM-3D²⁷. This method is based on sequential 2D DICOM images and utilizes 3D reconstruction and visualization technology to generate a stereoscopic 3D image with intuitive effects, which was a comparatively straightforward and highly efficient approach. We transformed the serial IF images into DICOM format and subsequently reconstruct it, and the same conclusion can be drawn, namely, PDGFR- α ⁺ cells almost constituted the whole structure of pulp and PDL, with NFATc1⁺ cells as subpopulation (Figure S6).

The previous sequencing analyses have reported the expression of NFATc1 in mandible and periodontal tissues²⁰. However, this evidence is limited to RNA-level sequencing, lacking in vivo data support. Therefore, with assistance of multiple genetic recombination systems, we next examined the expression pattern of NFATc1 in dental and PDL region of mandible M1, simultaneously observed the distribution atlas of PDGFR- α ⁺ cells. As shown in Figure 4A, the pulse timing points were same as above. Figure 4C and Video S3 showed the contoured mandible M1, including pulp and PDL with virtual dentin shell (white) in the buccal view, coronal view, and radicular view of pulp and PDL, respectively. According to the 3D-reconstruction results of 88 consecutive slices, it can be concluded that PDGFR- α ⁺ cells almost constituted the whole structure of dental pulp and periodontal ligament, while NFATc1⁺ cells most distributed in the pulp angle in the dental pulp tissue, and also scattered distributed in PDL region (Figure 4C&E and Figure S7). As shown in Figure 4C and Figure S8, the images obtained by confocal imaging can clearly obtain the distribution of PDGFR- α ⁺ and NFATc1⁺ cells from the single cell level.

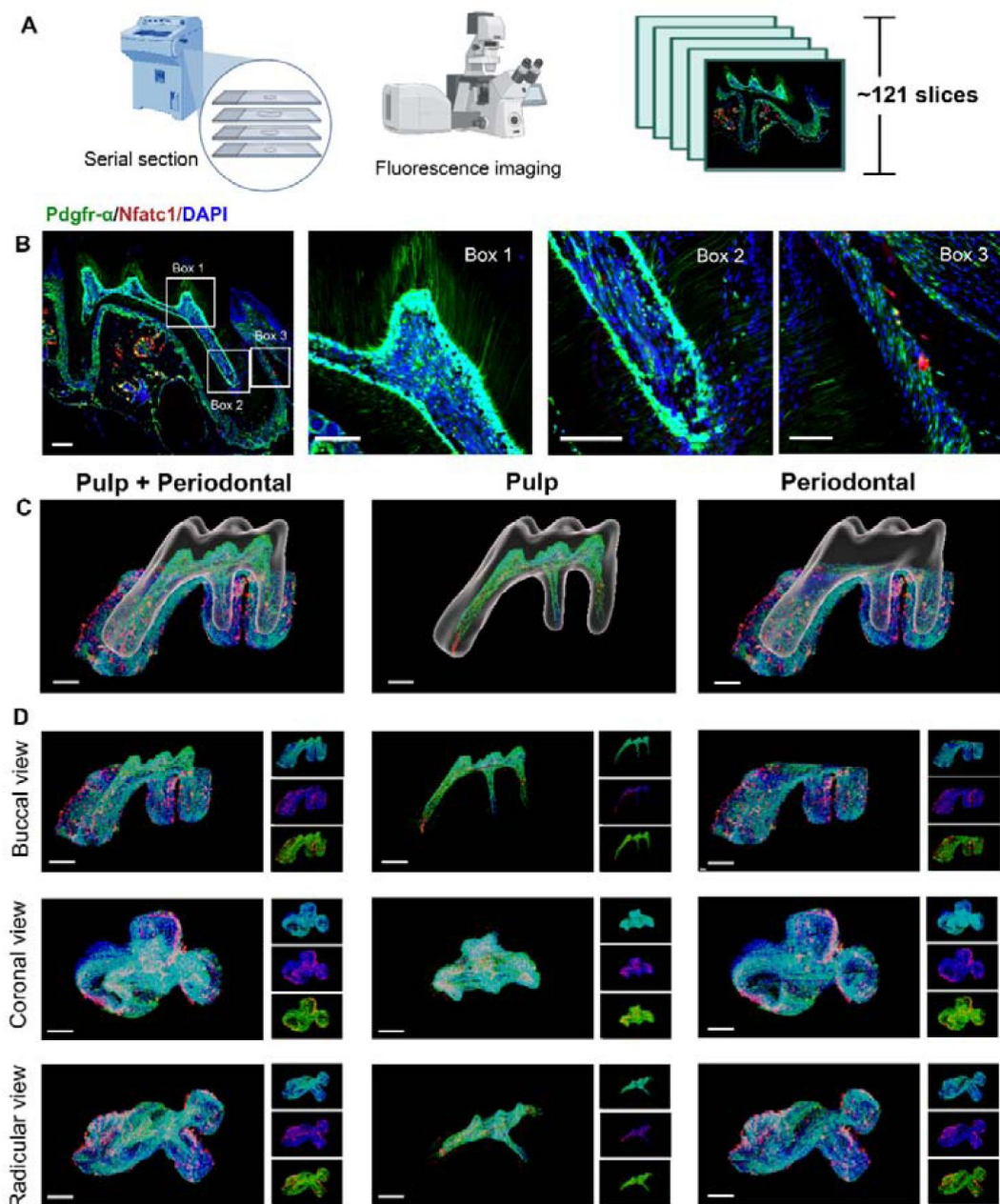


Figure 3.

Using traditional serial section for imaging of maxilla of *Pdgfr-α^{CreER}; Nfatc1^{DreER}; LGRT* mice (pulse). (A) Operation process of frozen section. Using this procedure, a total of 121 slices were collected in this sample. (B) Representative images acquired by confocal microscopy (scale bar = 200 μ m). Box 1: coronal pulp, Box 2: root pulp, Box 3: PDL, scale bar = 100 μ m. (C) Maxilla M1 after 3D reconstruction by imaris, including pulp and PDL with virtual dentin shell (white) in buccal view (scale bar = 300 μ m). (D) Image stack was displayed in buccal view, coronal view, and radicular view of pulp and PDL, respectively (scale bar = 300 μ m).

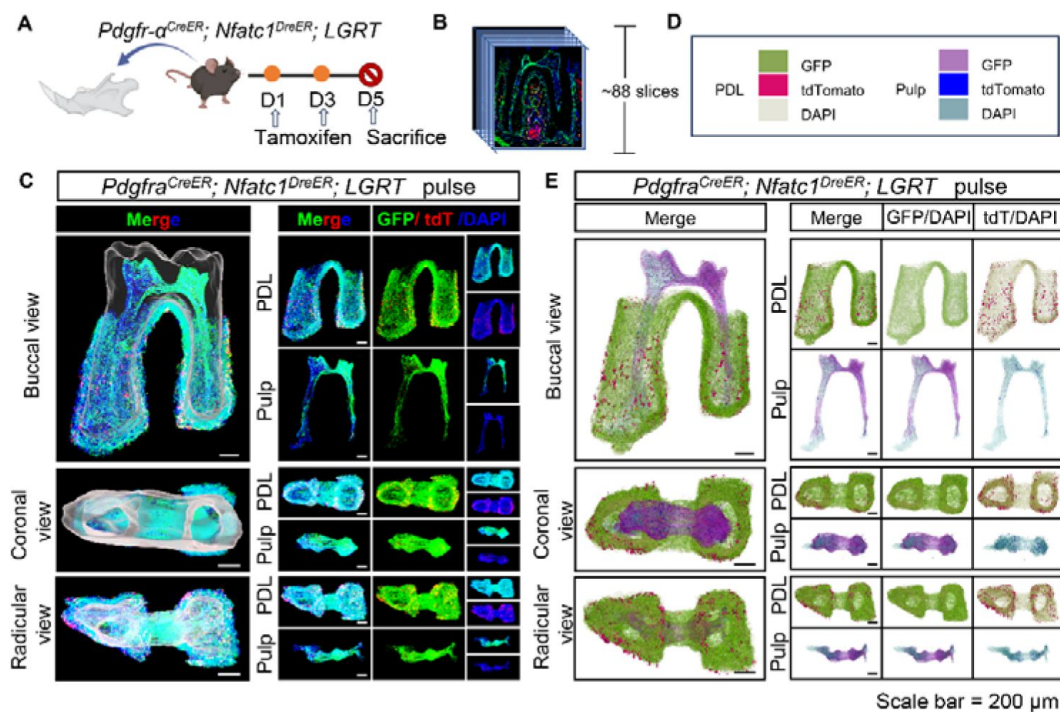


Figure 4.

Using traditional serial section for imaging of mandible of *Pdgfra*^{CreER}; *Nfatc1*^{DreER}; *LGRT* mice (pulse). (A) Schematic illustration of lineage tracing in *Pdgfra*^{CreER}; *Nfatc1*^{DreER}; *LGRT* mice. (B) Operation process of frozen section. Using this procedure, a total of 88 slices were collected in this sample. (C) Mandible M1 after 3D reconstruction by Imaris, including pulp and PDL with virtual dentin shell (white) in buccal view, coronal view and radicular view (scale bar = 200 μ m). (E) 3D reconstruction of mandible M1 by DICOM-3D; in PDL, ZsGreen+ cells in green, tdTomato+ cells in rose red; in pulp, ZsGreen+ cells in purple, tdTomato+ cells in blue. The image stack was also displayed in buccal view, coronal view, and radicular view of pulp and PDL, respectively. (scale bar = 200 μ m).

To further understand the presence of PDGFR- α ⁺ cells & NFATc1⁺ cells and their progeny cells after long-term tracking, we set a tracing timing points (**Figure 5A**). The results showed that nearly the whole pulp and PDL were composed of PDGFR- α ⁺ cells, and NFATc1⁺ cells were also scattered present in the root pulp and PDL region of maxilla M1 (**Figure 5B-D**, Video S4). Statistical data quantitatively showed that after 11 days of tracing, the number of PDGFR- α ⁺ cells, NFATc1⁺ cells, PDGFR- α ⁺NFATc1⁺ cells, and their progeny cells increased in both dental pulp and PDL (Figure S9). Specifically, the number of NFATc1⁺ cells in PDL area after tracing 11 days was 2.19-fold that of pulse NFATc1⁺ cells. Further, we also conducted cross-sectional analysis of the 3D reconstructed data, and selected the maximum cross-section of XZ axis which containing the whole pulp and PDL and the XY axis of the root pulp, respectively (**Figure 5E**). The results were similar to **Figure 2E**. From the cross section, only the Zs-Green fluorescence signal of PDGFR- α ⁺ cells can be observed and analyze while the tdTomato signal was illegible in target area, which was unable to distinguish from background interference signal. In the process of sample preparation, the soft tissue on the surface needs to be removed as much as possible before fixation, otherwise a similar situation will occur. However, for periosteum and other membranous structures that needed to be studied, the method of TC seems not suitable.

The tracing samples were also serially sliced and 3D reconstructed (**Figure 6** and **Figure 7**). A total of 117 slices were collected in maxilla of tracing sample (Figure S10) and a total of 120 slices were collected in mandible (Figure S15) from the jaw root PDL just appeared, to the buccal root PDL completely disappeared (10 μ m/slice). As shown in **Figure 6B**, Figure S11 and **Figure 7B**, Figure S16, the images obtained by confocal imaging can clearly obtain the distribution of PDGFR- α ⁺ and NFATc1⁺ cells from the single cell level. Consistent with the quantification of TC-based imaging results (Figure S9), the number of PDGFR- α ⁺ cells and NFATc1⁺ cells were significantly higher than that in pulse group, which indicated that there were a large number of PDGFR- α ⁺ & NFATc1⁺ cells and their progeny cells in crown pulp, root pulp and PDL after 11 days of tracing. In order to determine the existence of PDGFR- α ⁺NFATc1⁺ cells spatiotemporally, we also used software to 3D-reconstruct these 117 images. **Figure 6C-D** and Video S5 showed the contoured M1 of maxilla, including pulp and PDL, at buccal view, coronal view, and radicular view. **Figure 7C-D** and Video S6 showed the contoured mandible M1. Consistent with TC imaging and following quantitative results (Figure S9), the number of PDGFR- α ⁺NFATc1⁺ cells increased significantly in both dental pulp and periodontal ligament, similar to the phenomenon previously observed in long bone, which further encouraged us to follow up with this cell population, explore whether it is a specific mesenchymal stem cell population in periodontal tissue, and explore the indispensable role in periodontal homeostasis and regeneration. Worth notably, different from the whole imaging obtained by TC and light-sheet, the tdTomato signal reconstructed by traditional serial section-based confocal imaging method was very conspicuous, no matter in coronal pulp, root pulp, or PDL (Figure S12). We hypothesize that the current light-sheet systems for intact tissue-imaging have inherent limitations in capturing tdTomato signals, which become more evident in tissues with inherently low fluorescence strengths (in this work, due to the limitations of editing efficiency in DNA recombinase mediated lineage-tracing system, which guaranteed weaker tdTomato signal compared to ZsGreen). In contrast, traditional confocal imaging techniques do not encounter such issues.

However, due to the stratification of slices, the sample integrity was poor, leading to discontinuities in the z-axis. (Figure S13), which may render some biases in the results. We also utilized 3D reconstruction of mandible M1 by DICOM-3D (Figure S14). After 11 days tracing, the number of PDGFR- α ⁺ & NFATc1⁺ cells and PDGFR- α ⁺NFATc1⁺ cells increased significantly (**Figure 7**), which further supported our above conjecture that NFATc1⁺ cells were a subset of PDGFR- α ⁺ cells, and PDGFR- α ⁺NFATc1⁺ cells may contribute to mesenchymal stem cell population in dental and PDL regions.

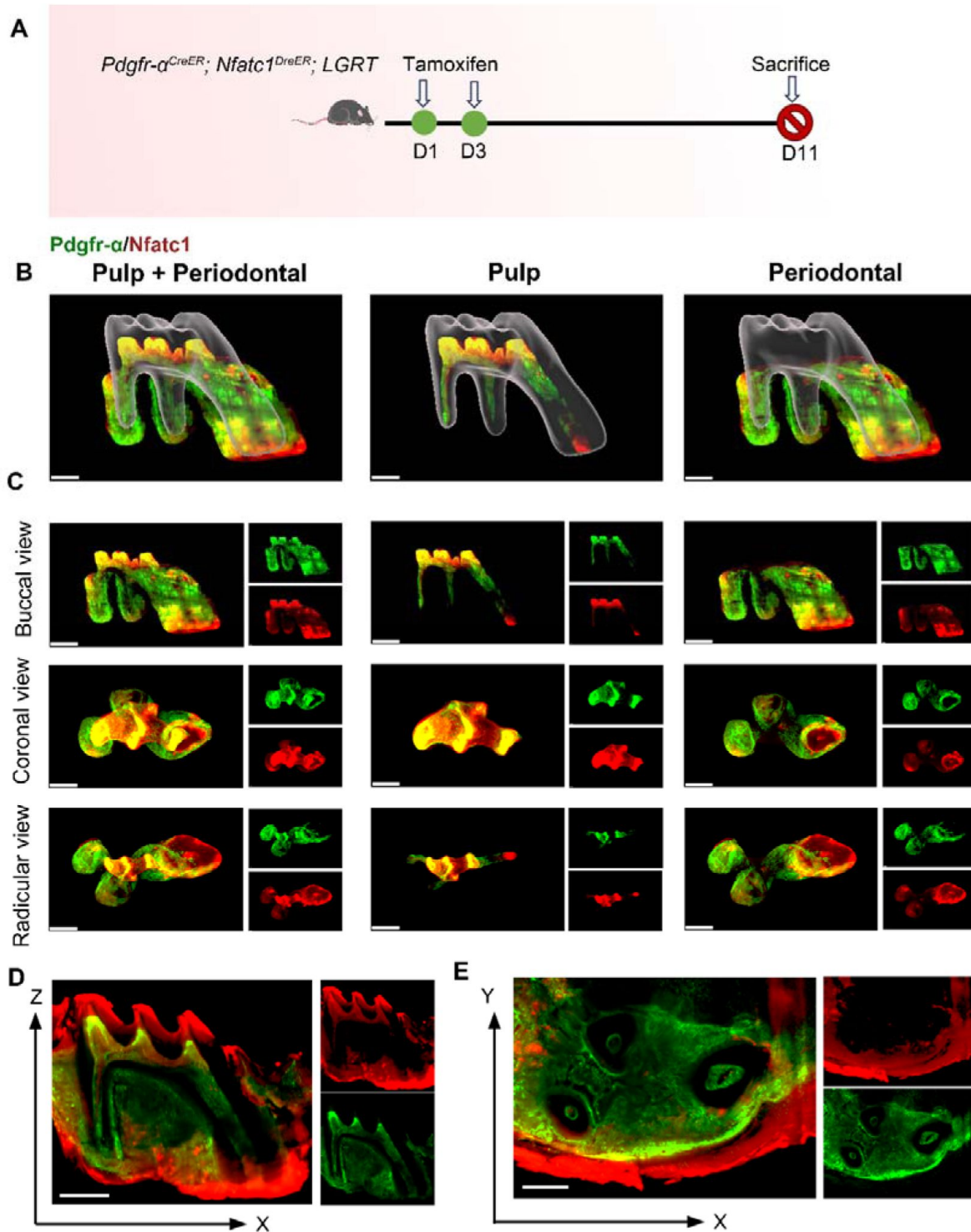


Figure 5.

The observation and analysis of lineage tracing of PDGFR- α^+ & NFATc1 $^+$ cells in M1 by whole-mount and high-speed imaging. (A) The flowchart of tracing. The mice were administrated with tamoxifen at D1 and D3, and sacrificed at D11. (B) The 3D images of contoured M1 of maxilla, including pulp and PDL with virtual dentin shell (white) in buccal view (scale bar = 300 μ m). (D) Image stack was displayed in buccal view, coronal view, and radicular view of pulp and PDL, respectively. (E) An optical slice was acquired on the X-Z (scale bar = 400 μ m) and X-Y direction to display the pulp and PDL (scale bar = 300 μ m).

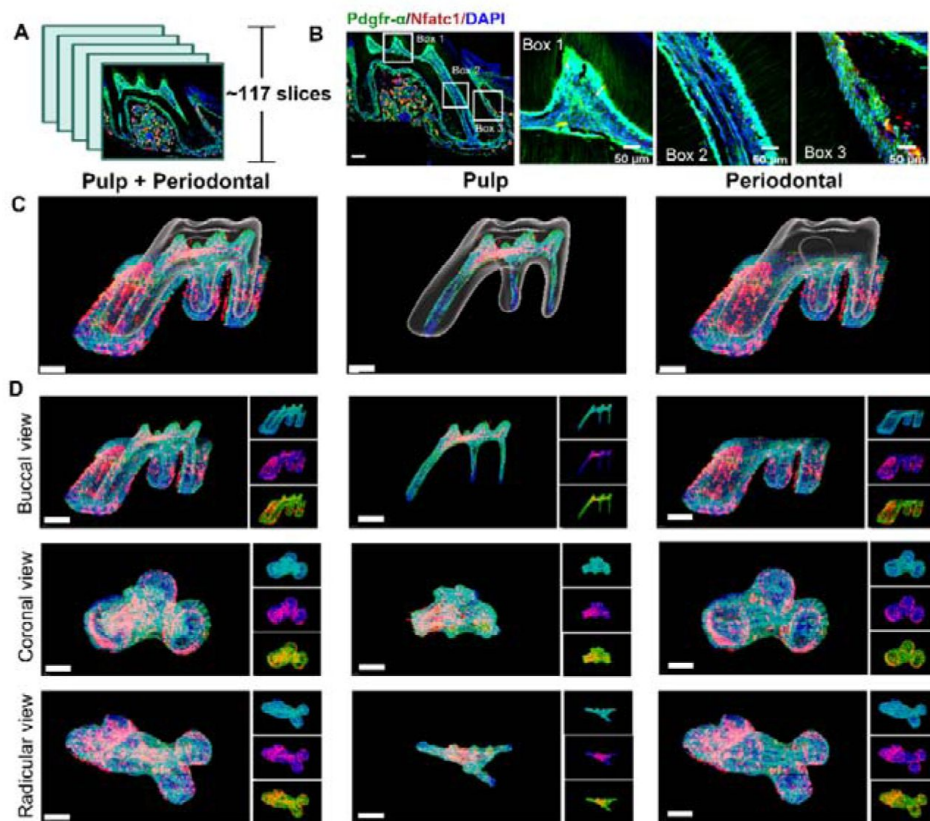


Figure 6.

Using traditional serial section for imaging of maxilla of *Pdgfr-α^{CreER}; Nfatc1^{DreER}; LGRT* mice (tracing 11 days). (A) Using this procedure, a total of 117 slices were collected in this sample. (B) Representative images acquired by confocal microscopy (scale bar = 200 μm). Box 1: coronal pulp; Box 2: root pulp; Box 3: PDL (scale bar = 50 μm). (C) Maxilla M1 after 3D reconstruction by imaris, including pulp and PDL with virtual dentin shell (white) in buccal view (scale bar = 300 μm). (D) Image stack was displayed in buccal view, coronal view, and radicular view of pulp and PDL, respectively (scale bar = 300 μm).

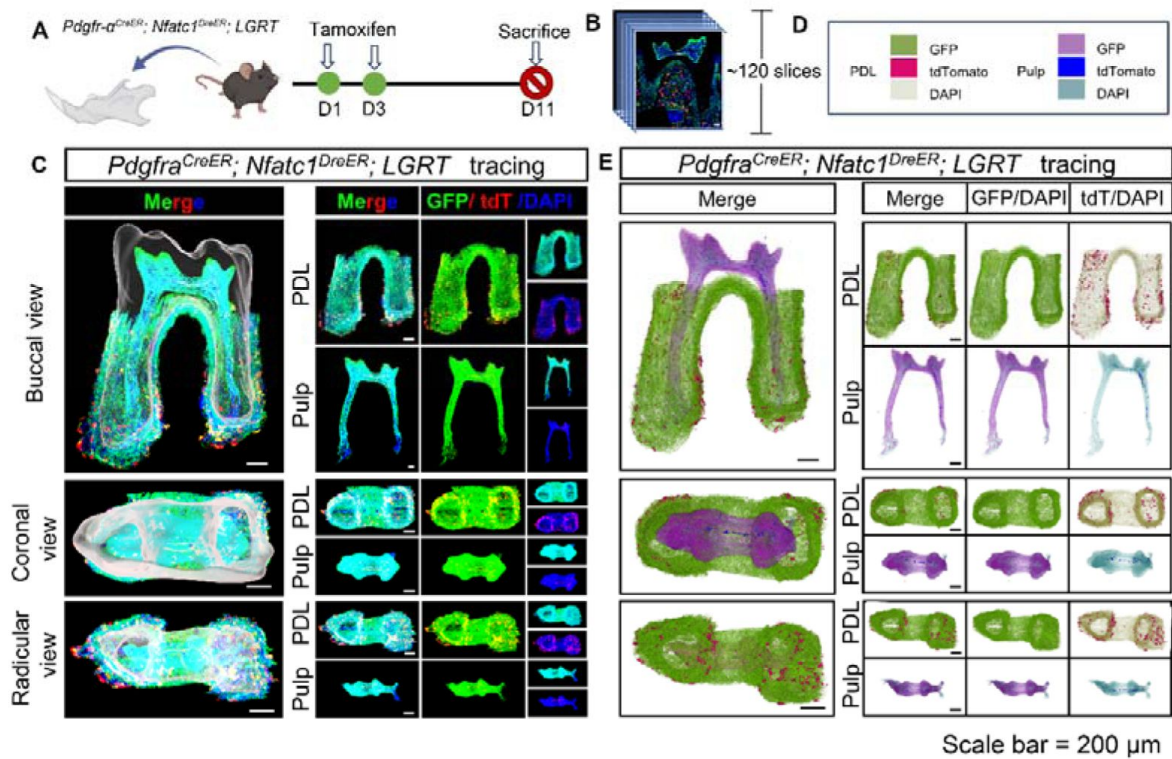


Figure 7.

H11-LGRT tracing distinct cell populations simultaneously in the pulp and PDL. (A) Schematic illustration of pulse and lineage tracing in *Pdgfr-α^{CreER}; Nfatc1^{DreER}; LGRT*. (B) 3D reconstruction of mandible M1 using Imaris, including pulp and PDL with virtual dentin shell (white) (scale bar = 200 μm). The image stack was displayed in buccal view, coronal view, and radicular view of pulp and PDL, respectively. (C) 3D reconstruction of mandible M1 using DICOM-3D, in which the distribution of NFATc1⁺ cells in pulp and PDL tissues can be more precisely perceive (scale bar = 200 μm).

2.2 Cranium and cranial sutures region

As a part of craniomaxillofacial hard tissue, we also intended to explore whether the presence of NFATc1⁺ and PDGFR- α ⁺ cells in cranial bone tissue/suture is different from dental and periodontal tissue (our previous study has identified the presence of NFATc1⁺ cells in the cranium by single-cell sequencing²⁸), therefore, we also obtained the cranium of *Pdgfr- α ^{CreER}; Nfatc1^{DreER}; LGRT* mice (pulse) for advanced TC imaging to familiarize NFATc1⁺/PDGFR- α ⁺ cells in cranium and cranial sutures region. As shown in Video S7-9, PDGFR- α ⁺ cells were widely distributed, including cranial sutures and bone interstitium, while NFATc1⁺ cells are mainly distributed in cranial sutures, partially coincided with the spatial location of PDGFR- α ⁺ cells. The PDGFR- α ⁺NFATc1⁺ cells located in the cranial suture were likely to participate in the physiological function of the cranial suture as a mesenchymal stem cell population.

The improved TC technology and following 3D imaging can deepen the understanding of specific cell populations in craniomaxillofacial tissues to a certain extent, and also lay a foundation for our future researches. However, it seemed not suitable for the study of periosteum, due to the strong auto-fluorescence under the light film system of soft tissues, which will affect the observation of target cells.

3. The hierarchical distribution of NFATc1⁺/PDGFR α ⁺ cells in IR1 lineage tracing system

From the above *Pdgfr- α ^{CreER}; Nfatc1^{DreER}; LGRT* system, the wider range of PDGFR- α ⁺ cells than NFATc1⁺ cells were observed, the wider range of PDGFR- α ⁺ cells than NFATc1⁺ cells were observed, we speculate that there may be a hierarchical relationship between the two. To further explore the relationship between PDGFR- α ⁺ and NFATc1⁺ cells in pulp and PDL and their hierarchical presence, respectively, we established the IR1 lineage tracing model in adult mice to assess the existence of these cells in the dental pulp and PDL. The structure of IR1 is *CAG-loxP-rox-Stop-loxP-ZsGreen-Stop-rox-tdTomato*, and the first *Cre-loxP* recombination would result in ZsGreen expression that removes a *rox* site, preventing the subsequent *Dre-rox* recombination in the same cell. The genotypes used were *PDGFR- α ^{CreER}; IR1; Nfatc1^{DreER}; IR1* and *PDGFR- α ^{CreER}; Nfatc1^{DreER}; IR1* (**Figure 8A-B**) in which NFATc1⁺ cells expressed tdTomato and PDGFR- α ⁺ cells were marked by ZsGreen. Tracing timing was set based on preliminary experimental outcomes. To comprehensively and accurately observe the distribution of NFATc1⁺ and PDGFR- α ⁺ cells in the pulp and PDL, we employed 3D reconstruction technology based on serial sections (**Figure 8C-E**, Video S10-12 and Figure S17-19, 21). As shown in **Figure 8C-E** and Figure S20, the images obtained by confocal imaging can clearly obtain the distribution of ZsGreen⁺ and tdTomato⁺ cells from the single cell level. In *Nfatc1^{DreER}; IR1* group, most tdTomato⁺ cells distributed in PDL region, while only sparse and punctate distribution in dental pulp, mostly in coronal pulp. These results were consistent with *Pdgfr- α ^{CreER}; Nfatc1^{DreER}; LGRT* mice. Moreover, since the recombinase recognition sites are interleaved (*loxP-rox-loxP-rox*), recombination by one system will naturally remove a recognition site of the other system, rendering its reporter gene inactive for further recombination. The results showed no tdTomato⁺ cells or ZsGreen⁺ cells were detected in the *Pdgfr- α ^{CreER}; IR1* or *Nfatc1^{DreER}; IR1* group respectively demonstrating the feasibility and accuracy of the IR1 system. In the *PDGFR- α ^{CreER}; Nfatc1^{DreER}; IR1* group, the distribution of Nfatc1⁺ cells in PDL were significantly fewer in number than in the *Nfatc1^{DreER}; IR1* group, while no significant difference in pulp area. In addition, PDGFR- α ⁺ cells were noted in both the dental pulp and PDL, with a wider distribution than NFATc1⁺ cells, predominantly in the odontoblastic layer of the pulp and also present in the pulp core. In the PDL, they were distributed from the gingival direction down to the root apex. These phenomena illustrated that PDGFR- α ⁺ cells were the precursor cell of NFATc1⁺ cells in PDL region. By contrast, there was almost no difference in

the number of NFATc1⁺ cells between *Nfatc1^{DreER}; IR1* group and *PDGFR-α^{CreER}; Nfatc1^{DreER}; IR1* group in dental pulp tissue, which was insufficient to certificate the hierarchical relationship of PDGFR-α⁺ and NFATc1⁺ cells in pulp area.

4. Identification of PDGFR-α⁺ and NFATc1⁺ population in pulp and PDL

4.1 Ablation of PDGFR-α⁺ cells disrupt the morphology of dental pulp and periodontal mesenchymal tissues

The results from the aforementioned lineage tracing experiments showed that PDGFR-α⁺ cells constitute a significant component of both dental pulp and periodontal tissues. Additionally, the hierarchical relationship experiments revealed that a portion of NFATc1⁺ cells in the periodontal ligament derives from PDGFR-α⁺ progenitor cells. Therefore, investigating the role of PDGFR-α⁺ cells in dental pulp and periodontal tissues has become more urgent. To explore the role of PDGFR-α⁺ cells in dental pulp and PDL tissues, we performed the cell ablation assay using *Pdgfr-α^{CreER}; DTA* mice with tamoxifen administered for 12 weeks (**Figure 9A** [↗](#)). The *Pdgfr-α^{CreER}; DTA* mice strain harbors a *loxP*-flanked termination cassette within the ubiquitous ROSA26 locus, controlling the expression of DTA. Tamoxifen administration was employed for in vivo toxin-mediated cell ablation (**Figure 9B** [↗](#)). Through H&E staining and Masson trichrome staining, we observed that in the control group, the odontoblast cell layer adjacent to the dentin consisted of densely arranged columnar cells both in the coronal or radicular pulp. However, this layer of odontoblasts was disrupted and nearly disappeared after ablation. Additionally, ablation of PDGFR-α⁺ cells in adult DTA mice also showed significant shrinkage in the central area of dental pulp tissue, specifically within the region of the pulp core, along with a substantial reduction in cell numbers (**Figure 9C-D** [↗](#)). H&E staining confirmed a decrease in periodontal composition following PDGFR-α⁺ cell clearance (**Figure 9E** [↗](#)). Furthermore, Masson's staining results revealed that the diminished portion primarily comprised periodontal fibrous tissue (**Figure 9F** [↗](#)). Therefore, it can be inferred that PDGFR-α⁺ cells predominantly constitute the interstitial components of both the periodontium and dental pulp. PDGFR-α⁺ cells within the dental pulp play a crucial role in the formation of the odontoblast cell layer and contribute significantly to most of the core tissue, while PDGFR-α is primarily involved in shaping periodontal components, particularly fibrous tissue.

4.2 PDGFR-α⁺ & NFATc1⁺ cells including MSCs and hematopoietic population in pulp and PDL

To identify the population of PDGFR-α⁺ and NFATc1⁺ co-expressing cells in the pulp and periodontal ligament (PDL), we generated *Pdgfr-α^{CreER}; Nfatc1^{DreER}; R26-LSL-RSR-tdT-DTR (LRTD)* mice. In these mice, Dre-rox and Cre-loxP recombination events enable the simultaneous expression of tdTomato and DTR specifically in pulp and PDL cells. In this experiment, we utilized the LRTD mice solely to label double-positive cells, so DTX was not administered to ablate the cells. To achieve high labeling efficiency, we treated the mice with tamoxifen twice (**Figure S22A**). Strong tdTomato signals were detected in both the PDL (**Figure S22B**) and pulp (**Figure S22C**). With respect to the MSC-specific marker AlphaV, we observed AlphaV⁺tdTomato⁺ cells in both regions. Additionally, CD45⁺ (hematopoietic marker) tdTomato⁺ cells were also present in these areas (**Figure S22B, C**). These findings suggest that the population of PDGFR-α⁺ and NFATc1⁺ co-expressing cells is heterogeneous.

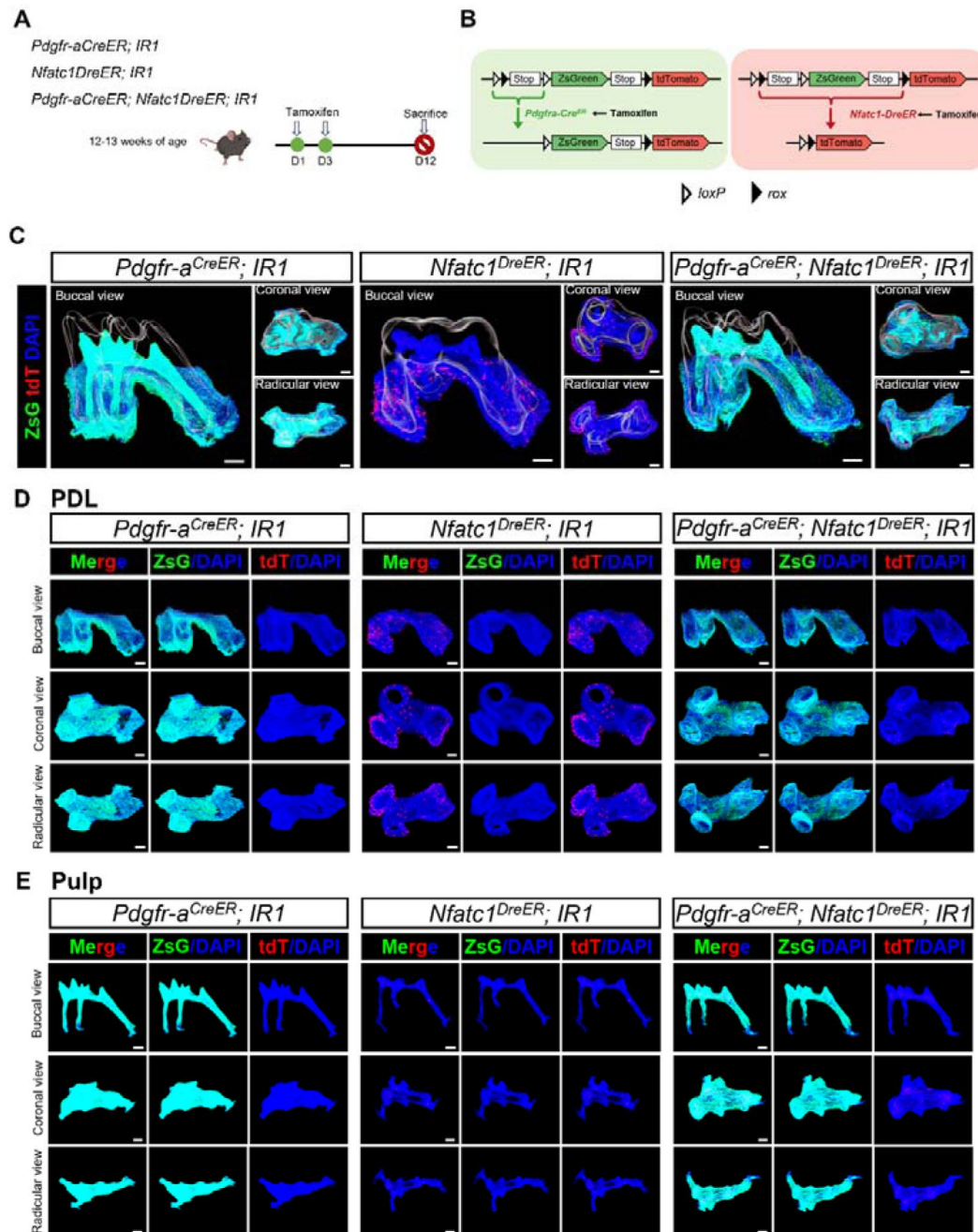


Figure 8.

IR1 tracing distinct cell populations simultaneously in the pulp and PDL. (A) Schematic illustration of lineaging tracing in *Pdgfr-α^{CreER}; IR1*, *Nfatc1^{DreER}; IR1* and *Pdgfr-α^{CreER}; Nfatc1^{DreER}; IR1* mice. (B) Schematic diagram of the IR1 working principle. (C) 3D reconstruction of maxilla M1 using Imaris, including pulp and PDL with virtual dentin shell (white) (scale bar = 200 μm). The image stack was displayed in buccal view, coronal view, and radicular view of PDL (D) and pulp (E), respectively.

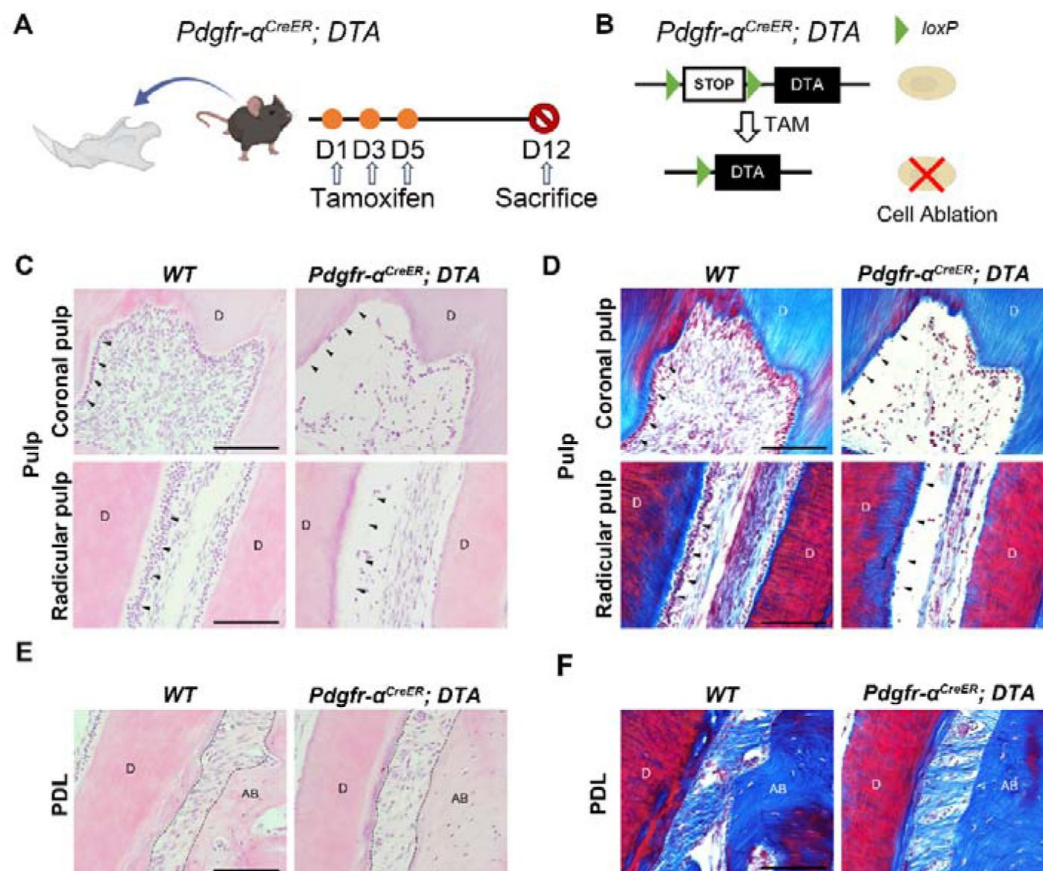


Figure 9.

Ablation of PDGFR- α^+ Cells Disrupts the morphology of dental pulp and periodontal ligament tissues. A) Schematics of tamoxifen induction. B) Schematic diagram of the DTA working principle. C, E) Representative H&E images of pulp C) and PDL E) of mandible M1 in *Pdgfr- α^{CreER} ; DTA* and control mice. D, F) Masson trichrome staining of pulp D) and PDL F) of mandible M1 in *Pdgfr- α^{CreER} ; DTA* and control mice. Arrows C, D) indicate the odontoblast cell layer. Dotted lines E, F) outline ROI of the PDL. D: dentin; AB: alveolar bone. Scale bar: 50 μ m.

Discussion

PDGFR- α and NFATc1, commonly acknowledged as markers for stem cells, are recognized for their contributions to stem cell characteristics. Our previous study also identified PDGFR- α and NFATc1 cells as SSCs in long bones. The conventional approach to studying the expression patterns of two cell populations involved techniques such as immunofluorescence or immunohistochemical staining, the generation of single-enzyme-cut editing gene mice, or a combination thereof. However, these traditional methods were often susceptible to factors like antibody potency, leading to incomplete and inaccurate depiction of the relationship between the two population of cells. In this study, we have developed two innovative genetic instruments, H11-LGRT and IR1, to facilitate distinct in vivo genetic lineage tracing assays. These tools were specifically designed for precision mapping of cellular lineages and have been utilized, for the first time, within the oral and maxillofacial regions. The arrangement of different recombination sites in relation to reporter genes is crucial for their use in in vivo tracking of multiple cell lineages. The ablation assay demonstrated that PDGFR- α nearly labeled almost mesenchyme cells, which was similar to results of H11-LGRT, noting that H11-LGRT is an available and reliable tool. Our results demonstrate that both IR1 and H11-LGRT are capable of tracing the cell fate mapping of various cells within the oral and craniofacial regions.

Dental pulp and the periodontal supporting tissues surrounding the tooth can be regarded as a relatively integrated biological functional unit. In the research of regeneration and repair mechanisms of dental pulp and PDL, the roles of periodontal and dental pulp stem cells are indispensable⁸. These cells play a vital role in the oral and maxillofacial regions, not only participating in routine tissue maintenance but also acting as key players in the self-repair process after injury^{29, 30}. Dental pulp stem cells have the potential to regenerate dental pulp tissue, while periodontal stem cells are crucial for the regeneration of periodontal tissues, including alveolar bone, PDL and gingiva^{31, 32}. The interactions between these cell types and their prospects in tissue engineering open new strategies and therapeutic approaches for modern dental treatments.

In our study, H11-LGRT systems have illuminated distinct expression patterns of PDGFR- α^+ , NFATc1 $^+$ and PDGFR- α^+ &NFATc1 $^+$ cells in PDL and dental pulp, indicating their diverse characteristics and roles. In PDL tissue, pulse data demonstrated widespread and abundant expression of PDGFR- α single-positive cells as well as NFATc1 single-positive cells, with no significant alteration in expression pattern or quantity after lineage tracing. Consequently, we conclude that in periodontal ligament and dental pulp tissues, PDGFR- α single-positive and NFATc1 single-positive cells primarily label intrinsic periodontal mesenchyme in PDL. Conversely, PDGFR- α^+ &NFATc1 $^+$ cells exhibited a more confined localization in PDL. The tracing data clearly illustrated that PDGFR- α^+ &NFATc1 $^+$ cells successfully gave rise to numerous progenies, which become predominant constituents within the periodontal ligament. In pulp tissue, the distribution of PDGFR- α single-positive cells was similar as that in PDL, primarily labeled odontoblast cell layer and there was not a significant increase in ZsGreen signal after tracing assay (**Figure. 5**). However, distinct outcomes were observed for NFATc1 $^+$ single-positive cells, as tracing experiments revealed an increased generation of progeny cells from those homogeneous NFATc1 within the intrinsic dental pulp. Consistently, the results for PDGFR- α^+ &NFATc1 $^+$ cells mirrored those of NFATc1 $^+$ single-positive cells. This observation may imply a niche-specific role for these cells especially PDGFR- α^+ &NFATc1 $^+$ cells in maintaining critical functions such as regeneration. This could indicate a more generalized role in tissue maintenance and repair, potentially contributing to the regeneration of both soft and hard tissues.

In order to investigate the hierarchical relationship between PDGFR- α^+ and NFATc1 $^+$ in the dental pulp and PDL, we developed IR1 system (**Figure. 8**). Our results revealed distinct hierarchical relationships in PDL and dental pulp. In periodontal tissues, compared to the control group (*Pdgfr- α^{CreER} ; IR1* and *Nfatc1 DreER ; IR1*), there was significantly less tdTomato $^+$ cells distribution in the *Pdgfr- α^{CreER} ; Nfatc1 DreER ; IR1* group while the signal of ZsGreen did not decrease significantly. Based on the function of the IR1 system, we could conclude that in the periodontal ligament, PDGFR- α^+ cells serve as precursor cells for some NFATc1 $^+$ cells. However, in dental pulp tissue, there was almost no difference in tdTomato $^+$ expression between the experimental and control groups, indicating that PDGFR- α^+ cells cannot be considered as precursor for NFATc1 $^+$ cells in the dental pulp, or there may not be a clear hierarchical relationship between the two. The sparse presence of NFATc1 $^+$ cells in both dental pulp and PDL, as observed in the *Nfatc1 DreER ; IR1* group, raised questions about the plasticity and differentiation potential of these cells. Are they primarily quiescent, being activated only upon injury, or do they continuously contribute to tissue homeostasis? Further studies, including functional assays and injury models, would be instrumental in clarifying the dynamic nature of these cells.

Cell ablation and immunofluorescence staining experiments further characterized the types and functions of PDGFR- α^+ /PDGFR- α^+ &NFATc1 $^+$ populations. After ablating PDGFR- α^+ cells, we observed damage to the odontoblast layer and shrinkage of the pulp core in dental pulp tissue, indicating that PDGFR- α^+ cells contribute to the composition of dental pulp tissue, particularly the odontoblast layer (**Figure. 9C, D**). In the periodontal ligament, we noted a reduction and destruction of collagen fibers, suggesting a role for PDGFR- α^+ cells in periodontal tissue structure (**Figure. 9E, F**). Previous results confirmed the presence of double-positive cells in both dental pulp and periodontal tissues and provided insights into their hierarchical relationships in the periodontal ligament (**Figure. 8**). To further investigate the double-positive cell population, we developed an inducible dual-editing enzyme reporter system to label these cells with tdTomato signals. Using AlphaV as a marker for mesenchymal stem cells (MSCs) and CD45 for hematopoietic cells, we found that double-positive cells included components of both MSCs and hematopoietic cells (Figure S22B, C), indicating a heterogeneous population. Further experiments are necessary to determine whether the predominant role in this co-positive MSC population is played by PDGFR- α^+ or NFATc1 $^+$ and to clarify the specific functions of these cells in the future.

For the traditional sectioning & confocal imaging procedure, it can obtain a highly distinct single-layer image. However, it is troubled to determine whether the different cell populations overlap in spatial location. At the same time, the experimental results were deeply affected by the differences between different section, which may cause error. It is also impossible to determine the amount and spatial position of a certain cell population in the whole tissue. In this study, we utilized two different techniques to map the distribution of aimed cells. We successfully achieved a 3D cellular-level visualization of PDL and pulp from *Pdgfr- α^{CreER} ; Nfatc1 DreER ; LGRT* mice by TC-based advanced imaging and 3D reconstructions based on continuous sections provide a more accurate representation of spatial distribution. TC-based advanced imaging procedure can clearly visualize its 3D structure, reconstruct the whole across latitudes, and understand the spatial position and expression of each structure, which could avoid the bias of traditional single-layer slicing may cause, and provides a more intuitive and objective description of the existing situation. However, our results demonstrated TC still has some limitations. During the experiment, we found that the signal of tdTomato in the pulp and PDL area was poor and could not be distinguished from autofluorescence through the imaging of light-sheet system, which may due to the limitations of editing efficiency in DNA recombinase-mediated lineage-tracing system, which guaranteed weaker tdTomato signal compared to ZsGreen. In addition, the total amount of NFATc1 $^+$ SSCs in these regions was limited, with the highly distribution in the alveolar bone, jaw bones, and trabecular bone surface edge, which guaranteed less visible signal in the target region. Another possibility is that the as-describe TC method may cause poor penetration of the tdTomato

fluorescence signals. The above results prompted that we should minimize the sample processing time to avoid fluorescence depletion to the greatest extent when using dual DNA recombinases-based genetic lineage tracing system for whole tissue imaging after TC treatment.

The 3D sections reconstruction results, however, effectively addressed the issue of weak tdTomato signal and provide a clearer visualization of the distribution of ZsGreen and tdTomato signals. For example, the tdTomato signal in the root pump, which was almost completely unobservable by TC-based imaging, can be clearly seen using confocal imaging and 3D reconstruction (**Figure 3C-D**, **Figure 6C-D**, and Figure S4, Figure S12). However, compared to TC, the quality of 3D reconstruction of sections still relies on the angle and quality of the sections, with the section angle having a significant impact on the reconstruction outcome. In addition, because the slice itself has a certain thickness (10 μ M in this study), which leads to the appearance of discontinuous in the final reconstructed image, and the aesthetics and accuracy could be affected to a certain extent. Also, unavoidable tissue damage during the sectioning process may result in the loss of some information. Therefore, a variety of different information could be obtained through two different imaging technologies, which prompt us to use the advanced experimental procedure according to the actual purpose.

The use of advanced genetic tools has also shed light on the limitations and challenges associated with current lineage tracing methodologies³³. For instance, the potential for recombinase leakiness or off-target effects necessitates the inclusion of stringent controls and corroborative experimental approaches³⁴. As such, the refinement of these tools and the development of additional specific markers are required to enhance the accuracy of cell fate mapping.

In conclusion, our study developed multiple DNA recombinases based on genetic lineage tracing and improved their accuracy and usability cooperating with advanced imaging techniques in order to map the atlas of NFATc1⁺/PDGFR- α ⁺ inclusive, exclusive and hierarchical distribution in dental and periodontal mesenchyme. Our study underscored the importance of NFATc1⁺ and PDGFR α ⁺ cells in the homeostasis and regeneration of oral and maxillofacial tissues. The nuanced understanding of their distribution and potential roles paves the way for novel regenerative strategies that could harness the intrinsic capabilities of these cells. Future research should focus on the functional characterization of these cells, their responsiveness to injury, and the interplay of signaling pathways that regulate their activity, which will be crucial for translating these findings into clinical applications.

Materials and methods

Methods

Ethics Statement and Animals

All animal procedures were reviewed and approved by Ethical Committees of West China School of Stomatology, Sichuan University (WCHSIRB-D-2017-041). Relative surgical models were conducted according to approved guidelines set by State Key Laboratory of Oral Diseases, West China Hospital of Stomatology. *Nfatc1*^{DreER} and *IR1* strain were kindly provided by Prof. Bin Zhou (Chinese Academy of Sciences), which were also reported in our previous study (Yu et al. 2022). *Pdgfr- α* ^{CreER} (Stock 018280) was purchased from the JAX Lab, *LSL-ZsGreen-RSR-tdTomato* (LGRT) was purchased from Shanghai Model Organisms Center (Strain# NM-KI-200319), and DTA mice were purchased from JAX Lab (Stock# 009669).

In vivo mouse studies

All the transgenic mice discussed in this work are adults around 12-14 weeks, which were intraperitoneally injected with tamoxifen (TAM) according to our previous report¹⁹.

RT-qPCR

For RT-qPCR total RNA was extracted using the TRIzolTM (Invitrogen) according to the manufacturer's protocol. Complementary DNA was synthesized by using the HiScript III RT SuperMix for qPCR (Vazyme) in accordance to user manuals. Then quantitative real-time PCR was performed in triplicate by using AceQ Universal SYBR qPCR Master Mix (Vazyme) for PCR reactions on an iCycler Real-Time Detection System (BioRad, USA).

Maxillae and cranium collection, fixation, and decalcification for light sheet imaging

Fresh dissected tissues were washed with PBS (Solarbio), and then rapidly transferred to 4% PFA (Sigma-Aldrich) for 4 hours. Before fixation, the muscle and fat attached to the tissue surface should be removed as much as possible, otherwise there will be strong interference signal during imaging. The tissues were then washed three times with PBS at 4 °C on a rocker platform for 15 min each time. For decalcification, tissues were incubated with 0.5 M EDTA solution (pH 7.4) at 4 °C for 24 h. After decalcification, the samples were washed three times with PBS on the rocker (15 min each time). The tissues were then submersed in 50%, 80% and 100% ethanol gradient for 30 min each to dehydrated. 100% ethanol was changed twice after every 20 min. Then the samples were immersed in 5% (V/V) H₂O₂ for 2 h (Sigma-Aldrich, H1009) for bleaching³⁵.

Clearing of maxillae and cranium

After bleaching, the samples were rinsed three times with Ethyl cinnamate (ECi) (Sigma-Aldrich, 112372) for 5 min each at room temperature.

Histological Preparation and Staining

Specifically, the tissues were fixed in 4% PFA at 4°C for 24 hours. After fixation, the maxilla was immersed in 30% sucrose solution at 4°C overnight. Then the samples were embedded in OCT tissue freezing medium (Leica) and placed on liquid nitrogen to conduct non-decalcified frozen sectioning at 7 to 10 µm thickness. The slices were treated with DAPI at room temperature for 1h. Finally, the slices were washed with PBS for 3 times, 10 minutes each time, and sealed for detection. For H&E staining and Masson trichrome staining, decalcified samples were dehydrated in graded ethanol and embedded in paraffin. Mandibles were sectioned into 6 µm slices using the microtome (Leica RM2255). H&E staining was performed according to the manufacturer's instruction (Biosharp). Masson trichrome staining was performed according to the manufacturer's instruction (Solarbio).

Imaging and 3D reconstruction of maxillae and mandible sections

Digital pathological system (Olympus FV3000) was used to scan all the stained sections and reconstructed by following the previously described protocol (Wu et al. 2020). The light sheet fluorescence microscopy-Zeiss Lightsheet Z.7 was used to acquire clearing samples microscopic image stacks.

Image analysis and quantifications

Slices/Z-stacks of images acquired on the light sheet and confocal microscope were processed and reconstructed in three dimensions with Imaris software (version 10.0.1). Imaris, Adobe Photoshop 25.0, and ImageJ were used for image processing and analysis in line with the journal's guidance for image processing. Quantification of cell numbers was done on a z stack of images in Imaris using the automatic spot detection feature.

3D surface reconstruction

3D surface rendering in images was applied using the surface module in Imaris.

Statistical analysis

All data were presented as the mean $\pm$ standard error of the mean (SEM). A two-tailed Student's t-test was used for comparison between groups. $P < 0.05$ was considered statistically significant. N.S. stands for not significant.

Impact statement

This work utilized multi-enzyme cutting lineage tracing system along with advanced imaging methods to systematically map the distribution atlas of NFATc1⁺ and PDGFR- α ⁺ cells in dental and periodontal mesenchyme.

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Conflict of Interests

All authors declare no conflicts of interest.

Author Contributions

F.Y. conceived the study; X.Y., C.H., C.Y., and F.L. performed research; X.Y., C.H., and F.L. analyzed data; X.Y. and C.H. wrote the manuscript; and L.Y., F.L. and F.Y. reviewed and edited the manuscript. X.Y contributed equally with C.H.

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Reviewer #1 (Public review):

Summary:

Utilizing transgenic lineage tracing techniques and tissue clearing-based advanced imaging and three-dimensional slices reconstruction, the authors comprehensively mapped the distribution atlas of NFATc1⁺ and PDGFR- α ⁺ cells in dental and periodontal mesenchyme and tracked their in vivo trajectories. This important work expands our understanding of both single and double positive NFATc1 and PDGFR- α cells in maintaining dental and periodontal mesenchyme homeostasis, and will provide impact on clinical application and investigation. The strength of this work is convincing, as it employed CRISPR/Cas9-mediated gene editing to generate two dual recombination systems, and mapped gNFATc1⁺ and PDGFR- α ⁺ cells residing in dental and periodontal mesenchyme, their capacity for progeny cell generation, and their inclusive, exclusive and hierarchical relations in homeostasis, generating a spatiotemporal atlas of these skeletal stem cell population.

This work has theoretical or practical implications in the periodontal field. The methods, data and analyses support the claims.

Comments on revised version:

The authors have addressed my main concerns.

<https://doi.org/10.7554/eLife.100173.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

In this study, Yang et al. investigated the locations and hierarchies of NFATc1+ and PDGFRα+ cells in dental and periodontal mesenchyme. By combining intersectional and exclusive reporters, they attempted to distinguish among NFATc1+PDGFRα+, NFATc1+PDGFRα-, and NFATc1- PDGFRα+ cells. Using tissue clearing and serial section-based 3D reconstruction, they mapped the distribution atlas of these cell populations. Through DTA-induced ablation of PDGFRα+ cells, they demonstrated the crucial role of PDGFRα+ cells in the formation of the odontoblast cell layer and periodontal components.

Thank you for your valuable comments and suggestions, which have greatly enhanced the quality of this research article. The manuscript has been significantly revised in accordance with the reviewers' comments. All necessary experimental conditions and required data have been included, and all the questions and considerations have been well-addressed in the revised manuscript and supporting information.

Main issues:

(1) The authors did not quantify the contribution of PDGFRα+ cells or NFATc1+ cells to dental and periodontal lineages in PDGFRαCreER; Nfatc1DreER; LGRT mice. Zsgreen+ cells represented PDGFRα+ cells and their lineages. Tomato+ cells represented NFATc1+ cells and their lineages. Tomato+Zsgreen+ cells represented NFATc1+PDGFRα+ cells and their lineages. Conducting immunostaining experiments with lineage markers is essential to determine the physiological contributions of these cells to dental and periodontal homeostasis.

Thanks for your question, we are sorry for the insufficient statement. Figure S9 provided statistical analysis of the number of PDGFR-α+ cells, NFATc1+ cells, and PDGFR-α&NFATc1+ cells in the dental pulp and periodontal ligament (PDL). The results allow for a clear comparison of the contributions of single-positive and double-positive cells to both tissues. Additionally, the tracing results showed whether these three cell populations have the capacity to produce progeny cells. We further supplemented the analysis with immunofluorescence results of double-positive cells to identify their cell types, selecting AlphaV as the marker for mesenchymal stem cells (MSCs) and CD45 as the marker for hematopoietic cells. This part is further discussed in the manuscript as below:

Page 14-15 in the revised manuscript, "To identify the population of PDGFR-α+ and NFATc1+ co-expressing cells in the pulp and periodontal ligament (PDL), we generated *Pdgfr-αCreER*; *Nfatc1DreER*; *R26-LSL-RSR-tdT-DTR (LRTD)* mice... Strong tdTomato signals were detected in

both the PDL (Figure S22B) and pulp (Figure S22C). With respect to the MSC-specific marker AlphaV, we observed AlphaV+tdTomato+ cells in both regions. Additionally, CD45+ (hematopoietic marker) tdTomato+ cells were also present in these areas (Figure S22B, C). These findings suggest that the population of PDGFR- α + and NFATc1+ co-expressing cells is heterogeneous.”

(2) The authors attempted to use PDGFRaCreER; Nfatc1DreER; IR1 mice to illustrate the hierarchies of NFATc1+ and PDGFRa+ cells. According to the principle of the IR1 reporter, it requires sequential induction of PDGFRa-CreER and Nfatc1-DreER to investigate their genetic relationship. Upon induction by tamoxifen, NFATc1+PDGFRa- cells and NFATc1-PDGFRa+ cells were labeled by Tomato and Zsreen, respectively. However, the reporter expression of NFATc1+PDGFRa+ cells was uncertain, most likely random. Therefore, the hierarchical relationship of NFATc1+ and PDGFRa+ cells cannot be reliably determined from PDGFRaCreER; Nfatc1DreER; IR1 mice.

Thank you for your question. We have supplemented the control group (*Pdgfr-aCreER; IR1*) experimental data (Figure 8). By comparing the results of *Pdgfr-aCreER; Nfatc1DreER; LGRT* tracing assays, we confirmed that the expression pattern and range of PDGFR- α + cells in pulp and PDL of *Pdgfr-aCreER; IR1* mice are consistent with those observed in *Pdgfr-aCreER; Nfatc1DreER; LGRT* mice (Figure 6), and the same applies to NFATc1+ cells. All of our experimental results have been repeated multiple times. In addition, the IR1 system was initially developed by Professor Bin Zhou's lab and was validated for feasibility and stability in a paper published in *Nature Medicine* in 2017 (<https://doi.org/10.1038/nm.4437>). Moreover, Professor Zhou Bo O's team applied *IR1* dual recombinases for bone lineage tracing in 2021 published in *Cell Stem Cell*, which also confirmed its feasibility and stability. (DOI: 10.1016/j.stem.2021.08.010)

Reviewer #2 (Public Review):

Summary:

Yang et al. present an article investigating the spatiotemporal atlas of NFATc1+ and PDGFR- α + cells within the dental and periodontal mesenchyme. The study explores their capacity for progeny cell generation and their relationships - both inclusive and hierarchical - under homeostatic conditions. Utilizing the Cre/loxP-Dre/Rox system to construct tool mice, combined with tissue transparency and continuous tissue slicing for 3D reconstruction, the researchers effectively mapped the distribution of NFATc1+ and PDGFR- α + cells. Additionally, in conjunction with DTA mice, the study provides preliminary validation of the impact of PDGFR- α + cells on dental pulp and periodontal tissues. Primarily, this study offers an in-situ distribution atlas for NFATc1+ and PDGFR- α + cells but provides limited information regarding their origin, fate differentiation, and functionality.

We would like to thank the reviewer for setting a high value on our study. Given many constructive suggestions, the manuscript has been revised to improve the quantity of this study. All the necessary discussions have also been added, and all the questions and concerns have been well-addressed in the revised manuscript. The point-to-point reply to the comments is listed below:

Strengths:

(1) Tissue transparency techniques and continuous tissue slicing for 3D reconstruction, combined with transgenic mice, provide high-quality images and rich, reliable data.

(2) The Cre/loxP and Dre/Rox systems used by the researchers are powerful and innovative.

(3) The IR1 lineage tracing model is significantly important for investigating cellular differentiation pathways.

(4) This study provides effective spatial distribution information of NFATc1+/PDGFR-α+ cell populations in the dental and periodontal tissues of adult mice.

Weaknesses:

(1) In the functional experiment section, the investigation into the role of NFATc1+/PDGFR-α+ cell populations is somewhat lacking.

Thank you so much for your comments and suggestions. We have supplemented the analysis with immunofluorescence results of double-positive cells to identify NFATc1+&PDGFR-α+ cell populations, selecting AlphaV as the marker for mesenchymal stem cells (MSCs) and CD45 as the marker for hematopoietic cells. This part was shown as below:

Page 14-15 in the revised manuscript, “To identify the population of PDGFR-α+ and NFATc1+ co-expressing cells in the pulp and periodontal ligament (PDL), we generated *Pdgfr-αCreER; Nfatc1DreER; R26-LSL-RSR-tdT-DTR (LRTD)* mice... Strong tdTomato signals were detected in both the PDL (Figure S22B) and pulp (Figure S22C). With respect to the MSC-specific marker AlphaV, we observed AlphaV+tdTomato+ cells in both regions. Additionally, CD45+ (hematopoietic marker) tdTomato+ cells were also present in these areas (Figure S22B, C). These findings suggested that the population of PDGFR-α+ and NFATc1+ co-expressing cells is heterogeneous.”

We also supplemented the discussion regarding the role of PDGFR-α+ population on page 17. Its potential role in pulp and periodontal formation had been suggested as well.

Page 17 in the revised manuscript, “After ablating PDGFR-α+ cells, we observed damage to the odontoblast layer and shrinkage of the pulp core in dental pulp tissue, indicating that PDGFR-α+ cells contribute to the composition of dental pulp tissue, particularly the odontoblast layer (Figure. 9C, D). In the periodontal ligament, we noted a reduction and destruction of collagen fibers, suggesting a role for PDGFR-α+ cells in periodontal tissue structure (Figure. 9E, F).”

(2) The author mentions that 3D reconstruction of consecutive tissue slices can provide more detailed information on cell distribution, so what is the significance of using tissue-clearing techniques in this article?

Thank you for your insightful comment, and we are sorry for the insufficient statement here. In our study, the utilization of tissue clearing techniques was to address some of the shortcomings associated with the 3D reconstruction of consecutive tissue slices, such as the compromised integrity of samples due to section layering, leading to discontinuities along the z-axis and potential loss of positive signals (Fig. S5, S13). Additionally, unavoidable tissue damage during the sectioning process may result in the loss of some information. As one of the most advanced imaging technologies currently available, tissue clearing/imaging allows for direct observation of the spatial location and relationships of fluorescently labeled cells within the intact tissue, which is more persuasive. Also, evolving beyond the analysis of structural and molecular biology of selected tissue sections, and expanding the focus to entire organs and organisms, is a trend in the development of the biomedical field (Nat Methods. 2024 Jul;21(7):1153-1165; Nat Commun. 2024 Feb 26;15(1):1764). Admittedly, no method is flawless; thus, our employment of two advanced imaging approaches aims to answer questions regarding the spatial positioning and relationships of PDGFR-α single-positive,

NFATc1 single-positive cells, and PDGFR- α ⁺ NFATc1⁺ cells from multiple perspectives. This is done to enhance the credibility and persuasiveness of our results.

We greatly appreciate your suggestion, which have significantly complemented the content of our article. The corresponding statements have been added in the revised manuscript as below:

Page 6 in the revised manuscript, “As one of the most advanced imaging technologies currently available, tissue clearing/imaging allows for direct observation of the spatial location and relationships of fluorescently labeled cells within the intact tissue. Therefore, according to the existing SUMIC tissue deep clearing (TC) methods, we modified and improved a rapid and efficient procedure, which enable rapid single-cell resolution and quantitative panoptic 3D light-sheet imaging.”

(3) After reading the entire article, it is confusing whether the purpose of the article is to explore the distribution and function of NFATc1⁺/PDGFR- α ⁺ cells in teeth and periodontal tissues, or to compare the differences between tissue clearing techniques and 3D reconstruction of continuous histological slices using NFATc1⁺/PDGFR- α ⁺ cells?

We sincerely appreciate your question and apologize for any ambiguous descriptions.

The purpose of our study is to map the atlas of NFATc1⁺/PDGFR- α ⁺ inclusive, exclusive and hierarchical distribution in dental and periodontal mesenchyme. Under this premise, the two advanced imaging techniques were merely employed as means to elucidate this issue. Indeed, in the previous manuscript, we did overemphasize the comparison and description of the differences between tissue clearing techniques and 3D reconstruction of continuous slices, which led to unnecessary misunderstandings for which we are deeply apologetic. Consequently, in this version of the manuscript, we have diminished the descriptions comparing their advantages and disadvantages, focusing instead on exploring the importance of NFATc1⁺/PDGFR- α ⁺ cells. We appreciate your suggestions once again.

Page 6 in the revised manuscript, “These two 3D-reconstruction and imaging technologies complement each other to jointly address the spatial positioning and hierarchical relationships of PDGFR- α ⁺, NFATc1⁺, and PDGFR- α ⁺ NFATc1⁺ cells from multiple perspectives.”

(4) The researchers did not provide a clear definition of the cell types of NFATc1⁺/PDGFR- α ⁺ cells in teeth and periodontal tissues.

Thanks for your suggestions. We discovered through cell ablation experiments that the removal of PDGFR- α ⁺ cells resulted in the destruction of the odontoblast layer in the dental pulp, shrinkage of the pulp core, and disruption of collagen fibers in the periodontal ligament. Combined with the results from lineage tracing, we conclude that PDGFR- α ⁺ cells primarily constitute the mesenchymal cells that form the supporting tissues in both the dental pulp and periodontal ligament (Part 4.1). Through immunofluorescence staining, AlphaV was as the marker for mesenchymal stem cells (MSCs) and CD45 as the marker for hematopoietic cells, we observed that the double-positive cell population was a heterogeneous group, containing both mesenchymal stem cells (MSC) and hematopoietic cells (Part 4.2).

(5) In studies related to long bones, the author defines the NFATc1⁺/PDGFR- α ⁺ cell population as SSCs, which as a stem cell group should play an important role in tooth development or injury repair. However, the distribution patterns and functions of the NFATc1⁺/PDGFR- α ⁺ cell population in these two conditions have not been discussed in this study.

Thanks for your suggestions. The NFATc1+/PDGFR-α+ cell population was identified as playing an important role in tissue regeneration, especially in oral and maxillofacial tissues. Our research primarily focuses on the identification of NFATc1+ and PDGFR-α+ cells within dental and periodontal mesenchyme, highlighting their contribution to tissue homeostasis and regeneration. Although the NFATc1+/PDGFR-α+ cells were characterized in the context of other tissue types, their detailed role in tooth development and injury repair remains an area for further exploration.

This part was further discussed on page 17-18 in the revised manuscript, “Cell ablation and immunofluorescence staining experiments further characterized the types and functions of PDGFR-α+/PDGFR-α+&NFATc1+ populations. After ablating PDGFR-α+ cells, we observed damage to the odontoblast layer and shrinkage of the pulp core in dental pulp tissue, indicating that PDGFR-α+ cells contribute to the composition of dental pulp tissue, particularly the odontoblast layer (Figure. 9C, D). In the periodontal ligament, we noted a reduction and destruction of collagen fibers, suggesting a role for PDGFR-α+ cells in periodontal tissue structure (Figure. 9E, F). Previous results confirmed the presence of double-positive cells in both dental pulp and periodontal tissues and provided insights into their hierarchical relationships in the periodontal ligament (Figure. 8). To further investigate the double-positive cell population, we developed an inducible dual-editing enzyme reporter system to label these cells with tdTomato signals. Using AlphaV as a marker for mesenchymal stem cells (MSCs) and CD45 for hematopoietic cells, we found that double-positive cells included components of both MSCs and hematopoietic cells (Figure S22B, C), indicating a heterogeneous population. Further experiments are necessary to determine whether the predominant role in this co-positive MSC population is played by PDGFR-α+ or NFATc1+ and to clarify the specific functions of these cells in the future.”

Reviewer #3 (Public Review):

Summary:

This groundbreaking study provided the most advanced transgenic lineage tracing and advanced imaging techniques in deciphering dental/periodontal mesenchyme cells. In this study, authors utilized CRISPR/Cas9-mediated transgenic lineage tracing techniques to concurrently demonstrate the inclusive, exclusive, and hierarchical distributions of NFATc1+ and PDGFR-α+ cells and their lineage commitment in dental and periodontal mesenchyme.

Strengths:

In cooperating with tissue clearing-based advanced imaging and three-dimensional slices reconstruction, the distribution and hierarchical relationship of NFATc1+ and PDGFR-α+ cells and progeny cells plainly emerged, which undoubtedly broadens our understanding of their in vivo fate trajectories in craniomaxillofacial tissue. Also, the experiment design is comprehensive and well-executed, and the results are convincing and compelling.

Weaknesses:

Minor modifications could be made to the paper, including more details on the advantages of the methodology used by the authors in this study, compared to other studies.

Thanks for your constructive comments and advice on how to improve the quality of this research article. We have thoroughly and carefully corrected the manuscript based on your suggestion, and all the necessary data have been added to support our claims. Meanwhile, all the questions and concerns have been well-addressed in the revised manuscript and the

revised supplementary information. Thus, we believe that the quality of this paper has been significantly enhanced. We thank you again for your great efforts.

Recommendations For The Authors:

Reviewer #1 (Recommendations For The Authors):

(1) Line 134, the authors categorized the reporter systems into three types: intersectional reporters, exclusive reporters, and nested reporters. However, Figure 1A does not depict the nested reporters.

Thanks for your helpful recommendation to improve the quality of this manuscript, and we are sorry for the mistake. In this revised manuscript, we have modified the content of Figure 1A, as displayed below:

(2) Line 238, the authors mentioned that NFATc1 is expressed in the mandible and periodontal tissues based on their previous sequencing analyses. It would be better to cite the related reference or display the expression of NFATc1 in the Supplemental Figures.

Thanks for your suggestions. We sincerely apologize for the typo that occurred during the writing process and have revised the original text to on page 9:

“The previous sequencing analyses have reported the expression of NFATc1 in mandible and periodontal tissues²⁰. (DOI: 10.1177/00220345221074356)”

(3) Line 264, the figure callout "Figure 5E" does not exist, and the figure legends of Figure 5 contain the same error.

We greatly appreciate your rigor and diligence, and we have corrected this error.

(4) Line 280, the figure callout "Figure S12" is incorrect.

Thank you for your efforts, and we are sorry for our negligence. The corresponding descriptions have been amended as below:

Page 10 in the revised manuscript, “Consistent with the quantification of TC-based imaging results (Figure S9), the number of PDGFR- α + cells and NFATc1+ cells were significantly higher than that in pulse group.”

(5) Line 301, the figure callout "Figure 4" is erroneous.

Thank you for your efforts, and we are sorry for our negligence. The corresponding descriptions have been amended as below:

Page 11 in the revised manuscript, “After 11 days tracing, the number of PDGFR- α + & NFATc1+ cells and PDGFR- α +NFATc1+ cells increased significantly (Figure 7)...”

(6) Line 306, the sentence "Our previous study identified the presence of NFATc1+ cells in the cranium by single-cell sequencing (unpublished data)" could be improved by referencing specific data or findings.

Thanks for your suggestions, and we are sorry for our negligence. The corresponding citation have been amended as below:

Page 11 in the revised manuscript, “As a part of craniomaxillofacial hard tissue, we also intended to explore whether the presence of NFATc1+ and PDGFR-α+ cells in cranial bone tissue/suture is different from dental and periodontal tissue (our previous study has identified the presence of NFATc1+ cells in the cranium by single-cell sequencing28”

(7) Line 341, the statement “Moreover, no PDGFR-α+ cells were detected in the *Nfatc1DreER; IR1* group,” needs further explanation or context.

Thanks for your suggestions. The corresponding descriptions have been amended as below:

Page 13 in the revised manuscript, “Moreover, since the recombinase recognition sites are interleaved (*loxP-rox-loxP-rox*), recombination by one system will naturally remove a recognition site of the other system, rendering its reporter gene inactive for further recombination. The results showed no tdTomato+ cells or ZsGreen+ cells were detected in the *Pdgfr-αCreER; IR1* or *Nfatc1DreER; IR1* group respectively demonstrating the feasibility and accuracy of the *IR1* system.”

(8) Several statements in this text were duplicated. For instance, lines 365 to 376 are identical to lines 497 to 508. This redundancy should be addressed to improve the manuscript's clarity and conciseness.

We greatly appreciate your suggestions, and we are sorry for the misunderstanding we may have caused. We have revised and integrated the entire Results 4 section (including lines 365 to 376 of the original manuscript) into the Discussion section to avoid unnecessary redundancy and misunderstandings. This adjustment also emphasizes that the goal of using two imaging techniques is to draw more credible conclusions from multiple perspectives, thereby mitigating the shortcomings of relying solely on existing advanced imaging methods. The revised content are as follows:

Page 18 in the revised manuscript, “TC-based advanced imaging procedure can clearly visualize its 3D structure, reconstruct the whole across latitudes, and understand the spatial position and expression of each structure, which could avoid the bias of traditional single-layer slicing may cause, and provides a more intuitive and objective description of the existing situation. However, our results demonstrated TC still has some limitations...”

Page 19 in the revised manuscript, “The 3D sections reconstruction results, however, effectively addressed the issue of weak tdTomato signal and provide a clearer visualization of the distribution of ZsGreen and tdTomato signals. For example, the tdTomato signal in the root pump, which was almost completely unobservable by TC-based imaging, can be clearly seen using confocal imaging and 3D reconstruction (Figure 3C-D, Figure 6C-D, and Figure S4, Figure S12). However, compared to TC, the quality of 3D reconstruction of sections still relies on the angle and quality of the sections, with the section angle having a significant impact on the reconstruction outcome. In addition, because the slice itself has a certain thickness (10 μM in this study), which leads to the appearance of discontinuous in the final reconstructed image, and the aesthetics and accuracy could be affected to a certain extent. Also, unavoidable tissue damage during the sectioning process may result in the loss of some information. Therefore, a variety of different information could be obtained through two different imaging technologies, which prompt us to use the advanced experimental procedure according to the actual purpose.”

Reviewer #2 (Recommendations For The Authors):

(1) It should be further highlighted in the article what cell type the NFATc1+/PDGFR-α+ cells should be defined as in teeth and periodontal tissues.

Thank you so much for your suggestions. We have supplemented the analysis with immunofluorescence results of double-positive cells to identify NFATc1+&PDGFR- α cell populations, selecting AlphaV as the marker for mesenchymal stem cells (MSCs) and CD45 as the marker for hematopoietic cells.

This part was on page 14-15 in the revised manuscript, “To identify the population of PDGFR- α and NFATc1+ co-expressing cells in the pulp and periodontal ligament (PDL), we generated *Pdgfr-aCreER; Nfatc1DreER; R26-LSL-RSR-tdT-DTR (LRTD)* mice... Strong tdTomato signals were detected in both the PDL (Figure S22B) and pulp (Figure S22C). With respect to the MSC-specific marker AlphaV, we observed AlphaV+tdTomato+ cells in both regions. Additionally, CD45+ (hematopoietic marker) tdTomato+ cells were also present in these areas (Figure S22B, C). These findings suggested that the population of PDGFR- α and NFATc1+ co-expressing cells is heterogeneous.”

We also supplemented the discussion regarding the role of PDGFR- α population on page 17. Its potential role in pulp and periodontal formation had been suggested as well:

Page 17 in the revised manuscript: “After ablating PDGFR- α cells, we observed damage to the odontoblast layer and shrinkage of the pulp core in dental pulp tissue, indicating that PDGFR- α cells contribute to the composition of dental pulp tissue, particularly the odontoblast layer (Figure. 9C, D). In the periodontal ligament, we noted a reduction and destruction of collagen fibers, suggesting a role for PDGFR- α cells in periodontal tissue structure (Figure. 9E, F).”

(2) The authors are advised to supplement the description of the cellular origin and the differentiation trajectory of NFATc1+/PDGFR- α cells in teeth and periodontal tissues.

Thank you for your suggestion. Our study currently focused more on mapping the distribution atlas of NFATc1+PDGFR α +, NFATc1+PDGFR α -, and NFATc1-PDGFR α cells in adult homeostatic mice. In the next step, we plan to explore the differentiation trajectory of NFATc1+/PDGFR α cells during development using single-cell sequencing and other methods.

(3) It is recommended to add figure labels to Figure 1B to facilitate reader comprehension.

Thank you for your valuable suggestion to improve the quality of this manuscript. We have modified Figure 1B in the revised manuscript as follows:

(4) Why compare 3D images from tissue clearing with 3D reconstructions of confocal imaging after consecutive tissue slicing?

Thanks for your important and helpful comments to improve the quality of this manuscript, and we are sorry for the insufficient statement.

The original intention of comparing the two methods was to draw more credible conclusions from multiple perspectives, thereby minimizing the limitations inherent in the singular use of current advanced imaging techniques. Indeed, the description in the previous manuscript could lead to misunderstandings among readers. Therefore, in the revised manuscript, we have modified and integrated the content of Results 4 section into the Discussion section to eliminate unnecessary verbosity and potential confusion.

Page 18 in the revised manuscript, “TC-based advanced imaging procedure can clearly visualize its 3D structure, reconstruct the whole across latitudes, and understand the spatial position and expression of each structure, which could avoid the bias of traditional single-layer slicing may cause, and provides a more intuitive and objective description of the existing situation. However, our results demonstrated TC still has some limitations...”

Page 19 in the revised manuscript, “The 3D sections reconstruction results, however, effectively addressed the issue of weak tdTomato signal and provide a clearer visualization of the distribution of ZsGreen and tdTomato signals. For example, the td-tomato signal in the root pump, which was almost completely unobservable by TC-based imaging, can be clearly seen using confocal imaging and 3D reconstruction (Figure 3C-D, Figure 6C-D, and Figure S4, Figure S12). However, compared to TC, the quality of 3D reconstruction of sections still relies on the angle and quality of the sections, with the section angle having a significant impact on the reconstruction outcome. In addition, because the slice itself has a certain thickness (10 μ M in this study), which leads to the appearance of discontinuous in the final reconstructed image, and the aesthetics and accuracy could be affected to a certain extent. Also, unavoidable tissue damage during the sectioning process may result in the loss of some information. Therefore, a variety of different information could be obtained through two different imaging technologies, which prompt us to use the advanced experimental procedure according to the actual purpose.”

(5) The experimental results section does not specify the age of the mice used, which lacks clarity for the reader and makes it difficult to determine at what developmental stage the observed distribution of NFATc1+/PDGFR- α + cells occurs.

Thank you for your suggestion. I apologize for overlooking this point. I only displayed the age of the mice in some of the figures. All the transgenic mice discussed in this article are adults around 12-14 weeks. I have added the specific weeks of age in the main text.

(6) What is the rationale behind selecting day 1, day 3, and day 5 as the experimental time points in Figure 2B?

Thanks for your questions. 48 hours after injection, TAM can be metabolized in the body and converted into 4-OHT, which then distributes thoroughly to various tissue systems through the bloodstream. Therefore, we chose to administer a booster dose 48 hours after the initial injection to ensure timely replenishment and achieve high labeling efficiency. This drug administration scheme has already been validated for feasibility in our preliminary studies.

(7) In Figure 2E, why is there a large area of red signal visible in the tooth enamel?

Thanks for your valuable comments and advice on how to improve the quality of this research article and our future work. As we discussed in the main text, the existing TC-based imaging techniques cannot meet the requirements for capturing as conspicuous tdTomato signals as ZsGreen, which may due to: 1) the editing efficiency of the DNA recombinase-mediated lineage-tracing system has limitations; 2) the lower presence of NFATc1+ cells in the region-of-interest (ROI) ensures weak signals of tdTomato; 3) the TC method as described may result in poor penetration of td-tomato fluorescence signals. Therefore, to clearly display the NFATc1+ cells in the ROI (periodontal ligament, pulp, and alveolar bone) as much as possible, we increased the intensity of excitation fluorescence of 561-channel of the Lightsheet fluorescence microscopy, which led to a large area of unrelated red signal in non-target areas (tooth enamel). In future work, we will further improve the TC procedure to shorten the sample processing time, and developing other transgenic mice to address this issue. Thanks again.

(8) In the text at Line 249, the author notes that PDGFR α + cells are widely distributed, and NFATc1+ cells are primarily located in the pulp horns. What is the relevance of their distribution to their function?

Thank you very much for your suggestion. We found that PDGFR α ⁺ cells are widely distributed in dental pulp tissue. Combined with the results from subsequent cell ablation experiments, it revealed that PDGFR α ⁺ cells contribute to the formation of the odontoblast layer and the pulp core. In our supplementary data, we discovered through immunofluorescence staining that double-positive cells co-expressed AlphaV in the dental pulp, indicating that they possessed MSC components. We need to further investigate the relationship between their distribution and function in the future.

(9) In Line 301 of the text, there is a mislabeling of Figure 4. Please verify this carefully throughout the document.

Thank you for your efforts, and we are sorry for our negligence. We have made the necessary corrections and have meticulously reviewed the entire manuscript to ensure that there were no similar mistakes. The corresponding descriptions have been amended as below:

Page 11 in the revised manuscript, “After 11 days tracing, the number of PDGFR- α ⁺ & NFATc1⁺ cells and PDGFR- α ⁺NFATc1⁺ cells increased significantly (Figure 7)...”

(10) Between Lines 323 to 325, the author states: "the wider range of PDGFR- α ⁺ cells than NFATc1⁺ cells were observed, which laid the foundation for our conjecture that NFATc1⁺ cells may contribute as subpopulation of PDGFR- α ⁺ cells." This statement is inaccurate.

Thank you for your suggestions. We apologize for the inaccuracies in our description and have made corrections in the original text.

Page 12 in the revised manuscript, “the wider range of PDGFR- α ⁺ cells than NFATc1⁺ cells were observed, we speculate that there may be a hierarchical relationship between the two.”

(11) The author is advised to combine the use of single-cell sequencing data for cell trajectory analysis to corroborate the differentiation relationships between NFATc1⁺/PDGFR- α ⁺ cells, discussing their specific origins and final differentiation fates.

Thank you for your suggestion; it is very meaningful to us and will be the focus of our future research work.

(12) In the Results 4 section, the comparison between tissue clearing imaging and 3D reconstruction of consecutive tissue slices could be discussed in the discussion section.

We greatly appreciate your suggestions. We have revised and integrated the entire Results 4 section into the Discussion section to avoid unnecessary redundancy and misunderstandings. This adjustment also emphasizes that the goal of using two imaging techniques is to draw more credible conclusions from multiple perspectives, thereby mitigating the shortcomings of relying solely on existing advanced imaging methods. The revised content are as follows:

Page 18 in the revised manuscript, “TC-based advanced imaging procedure can clearly visualize its 3D structure, reconstruct the whole across latitudes, and understand the spatial position and expression of each structure, which could avoid the bias of traditional single-layer slicing may cause, and provides a more intuitive and objective description of the existing situation. However, our results demonstrated TC still has some limitations...”

Page 19 in the revised manuscript, “The 3D sections reconstruction results, however, effectively addressed the issue of weak tdTomato signal and provide a clearer visualization of the distribution of ZsGreen and tdTomato signals. For example, the tdTomato signal in the root pump, which was almost completely unobservable by TC-based imaging, can be clearly seen using confocal imaging and 3D reconstruction (Figure 3C-D, Figure 6C-D, and Figure S4,

Figure S12). However, compared to TC, the quality of 3D reconstruction of sections still relies on the angle and quality of the sections, with the section angle having a significant impact on the reconstruction outcome. In addition, because the slice itself has a certain thickness (10 μM in this study), which leads to the appearance of discontinuous in the final reconstructed image, and the aesthetics and accuracy could be affected to a certain extent. Also, unavoidable tissue damage during the sectioning process may result in the loss of some information. Therefore, a variety of different information could be obtained through two different imaging technologies, which prompt us to use the advanced experimental procedure according to the actual purpose.”

(13) The article only demonstrates the impact of removing PDGFR- α cells on the dental pulp and periodontal tissues of adult mice. What would be the impact of removing NFATc1 α cells on teeth and periodontal tissues?

Thank you for your suggestions. Our lab had been investigating the role of NFATc1+ cells in PDL and dental pulp tissues which is currently submitted to another journal. So please forgive me for not being able to present the data. The ablation assays showed that NFATc1+ cells may be involved in the formation of the odontoblast layer in dental pulp and in promoting osteogenic differentiation in the periodontal ligament.

(14) The effects of removing PDGFR- α cells on the teeth and periodontal tissues of adult mice are shown in the article. What would be the impact on teeth and periodontal tissues if PDGFR- α cells were removed during early development?

Thank you for your question. Our current research has not yet focused on the impact of PDGFR- α cells on the formation of periodontal ligaments and dental pulp tissue during the developmental stage. In our literature search, we found articles indicating that PDGFR- α was expressed at all stages of tooth development, and that PDGFR- α signaling was crucial for regulating the growth of the tooth apex and the proper extension of the palatal shelves during palatal fusion. Disruption of PDGFR α signaling interferes with apex growth and the critical extension of palatal shelves during craniofacial development. In the future, we would like to focus on the role of PDGFR- α cells during teeth development.

(15) If the data on the skull are not presented in this paper, it is suggested not to overly describe it in the results section, or to include related skull data in supplementary figures.

We appreciate your attention to detail and your suggestions for improving the clarity and presentation of our work. The corresponding results of cranium and cranial sutures region were shown in Video S7-9 in the revised manuscript.

Reviewer #3 (Recommendations For The Authors):

We sincerely appreciate your thorough review and positive feedback on our manuscript. In accordance with your recommendations, all the questions and concerns have been well-addressed in the revised manuscript. We believe these revisions further enhance the clarity and quality of our work. The point-to-point reply to the comments is listed below:

(1) In line 181, the author claimed that "we modified and improved a rapid and efficient procedure...this ultrafast clearing technique could minimize the impact on transgenic mice." However, there is no mention in the main text of the amount of time required for other methods. How can the "rapid" element of your improved method be reflected? The author should briefly list a few other studies and discuss them.

Thanks for your important and helpful comments, and we are sorry for the insufficient statement. In recent years, a variety of tissue clearing methods have emerged. Here is a summary of the methods and durations used for hard tissue clearing as published in several authoritative journals:

Author response table 1.

Methods	Durations	References
MarShie clearing procedure	tissue 13-15 days	Nat Commun. 2024, 15, 1764
FDISCO procedure	clearing 4.5 days	Sci Adv. 2019, 5, eaau8355
PEGASOS method	12 days	Cell Research. 2018, 28, 803–818.
iDISCO	4.5 days	J Comp Neurol. 2024 532, e25582.

In comparison, our approach requires only approximately two days, thereby minimizing the potential damage to the tissue itself. Additionally, the study employs transgenic mice mediated by lineage tracing, and the shorter processing time also serves to reduce the impact on the fluorescence of the positive cells to a minimum.

(2) In Figure S6, the author mentioned the use of another 3D reconstruction method-DICOM-3D. What is the advantage of this methodology? Is the conclusion drawn the same as the previous approaches? The author should propose corresponding discussions in this section.

We sincerely appreciate your comments. The purpose of employing DICOM-3D reconstruction for the serial section images is to validate the constructed results obtained by Imaris. This method is based on sequential 2D DICOM images and utilizes 3D reconstruction and visualization technology to generate a stereoscopic 3D image with intuitive effects. Compared to Imaris reconstruction, this method offers a more straightforward and time-efficient approach. Regardless of the different reconstruction methods employed in this study, the ultimate goal remains consistent, which is to jointly address the spatial positioning and hierarchical relationships of PDGFR- α +, NFATc1+, and PDGFR- α NFATc1+ cells from multiple perspectives, to enhance the credibility and persuasiveness of our results. We have also included the corresponding description in the revised manuscript as follows:

Page 8-9 in the revised manuscript, “To enhance the comprehensive and accurate display of the reconstruction results and to mitigate the potential errors that may arise from relying on single reconstruction method, we employed an alternative 3D reconstruction method—DICOM-3D. This method is based on sequential 2D DICOM images and utilizes 3D reconstruction and visualization technology to generate a stereoscopic 3D image with intuitive effects, which was a comparatively straightforward and highly efficient approach. We transformed the serial IF images into DICOM format and subsequently reconstruct it, and the same conclusion can be drawn, namely, PDGFR- α + cells almost constituted the whole structure of pulp and PDL, with NFATc1+ cells as subpopulation (Figure S6).

(3) Line 292: Why was the tdTomato signal in confocal-based reconstruction more conspicuous than the TC procedure? Some descriptions would be beneficial for readers' understanding.

Thank you very much for your comments. We hypothesize that the current light-sheet systems have inherent limitations in capturing tdTomato signals of intact tissue, which become more evident in tissues with inherently low fluorescence strengths (in this work, due to the limitations of editing efficiency in DNA recombinase mediated lineage-tracing system,

which guaranteed weaker tdTomato signal compared to ZsGreen). In contrast, traditional confocal imaging techniques do not encounter such issues. The corresponding descriptions in the revised manuscript are shown as follows:

Page 11 in the revised manuscript, “We hypothesize that the current light-sheet systems for intact tissue-imaging have inherent limitations in capturing tdTomato signals, which become more evident in tissues with inherently low fluorescence strengths (in this work, due to the limitations of editing efficiency in DNA recombinase mediated lineage-tracing system, which guaranteed weaker tdTomato signal compared to ZsGreen). In contrast, traditional confocal imaging techniques do not encounter such issues.”

(4) Part 2.2, line 305: What is the purpose of analyzing the cranium and cranial sutures region through TC technology?

Thank you for your comments. There are three main purposes of this part of the experiment. First, our research group has long been committed to studying the distribution and role of NFATc1+ SSCs in a variety of hard tissues, and our previous study has identified the presence of NFATc1+ cells in the cranium by single-cell sequencing. Therefore, in this work, we also intend to investigate the spatiotemporal atlas of NFATc1+ and PDGFR-α+ cells in cranium and cranial sutures region based on transgenic lineage tracing techniques. Second, as a part of craniomaxillofacial hard tissue, we intended to explore whether the presence of NFATc1+ and PDGFR-α+ cells in cranial bone tissue/suture is different from dental and periodontal tissue; In addition, the results in Video S7-9 further demonstrated that our improved tissue clearing procedure in this work is universal for a variety of hard tissues, which lay a foundation for our future researches.

Page 11 in the revised manuscript, “As a part of craniomaxillofacial hard tissue, we also intended to explore whether the presence of NFATc1+ and PDGFR-α+ cells in cranial bone tissue/suture is different from dental and periodontal tissue (our previous study has identified the presence of NFATc1+ cells in the cranium by single-cell sequencing)”

(5) Some images before & after the tissue-clearing procedure need to be provided in the supplemental file.

Thanks for your important and helpful comments to improve the quality of this manuscript. We have included the corresponding description and photographs in the main text and the supplemental file as follows:

Page 7 in the revised manuscript, “As shown in Figure S1A-B, we recorded bright-field images of the maxilla before and after clearing, and our procedure achieved high transparency of the whole tissue. On this basis, whole-tissue imaging can be achieved, with the observation of different cell type distribution in spatial 3D structure.”

(6) In part 5, line 394, the author investigated the consequences of the ablation of PDGFR-α+ cells in dental pulp and periodontal mesenchymal tissues, but some research objectives and mechanisms need to be discussed here, regarding: "why choosing to ablation PDGFR-α+ cells instead of NFATc1+ cells? Was the hierarchical relationship between PDGFR-α+ cells and NFATc1+ cells considered during the experimental design?", etc.

Thank you very much for your suggestion, it has been very helpful. We chose PDGFR-α+ cells as the subject for the cell ablation experiments based on the results from the previous lineage tracing and hierarchical relationship studies. We have included the corresponding description and photographs in the main text and the supplemental file as follows:

Page 13 in the revised manuscript, “The results from the aforementioned lineage tracing experiments showed that PDGFR- α + cells constitute a significant component of both dental pulp and periodontal tissues. Additionally, the hierarchical relationship experiments revealed that a portion of NFATc1+ cells in the periodontal ligament derives from PDGFR- α + progenitor cells. Therefore, investigating the role of PDGFR α + cells in dental pulp and periodontal tissues has become more urgent.”

(7) Some claims in the main text were lack of literature citation, such as in lines 207 and 234.

Thank you very much for your comments. We are deeply sorry for the mistakes. We have added the relevant references at the appropriate locations in the main text as follows:

(1) line 207 of previous manuscript (page 8, line 206 in the revised manuscript): We sincerely apologize for the typo that occurred during the writing process and have revised the original text to: which was consistent with RNA-sequencing results in the previous study²⁰. (DOI: 10.1177/00220345221074356)

(2) line 234 of previous manuscript (page 9, line 234 in the revised manuscript): “we employed an alternative 3D reconstruction method—DICOM-3D²⁷.” (DOI: 10.1177/09544119211020148)

(8) What were the specific reasons for the conspicuous tdTomato signal in the reconstructed images obtained by traditional serial section-based confocal imaging, which were not as evident in TC imaging?

Thank you very much for your comments. Traditional sectioning and subsequent confocal imaging can clearly display fluorescence signals on a single plane (Figure 3B, Figure 6B, Figure S3, S8, S11, S16, S19), therefore, after 3D reconstruction of multiple planes, it will still have a high resolution (Figure 3, 4, 7, 8). However, for TC imaging, the current light-sheet systems have inherent limitations in capturing tdTomato signals of intact tissue, which become more evident in tissues with inherently low fluorescence strengths (in this work, due to the limitations of editing efficiency in DNA recombinase mediated lineage-tracing system, which guaranteed weaker tdTomato signal compared to ZsGreen). In contrast, traditional confocal imaging techniques do not encounter such issues.

(9) In tissue clearing techniques, do the chemical reagents and procedures used affect the signal intensity of tdTomato and ZsGreen?

We appreciate your helpful comment. In this work, we modified and improved a rapid and efficient tissue deep clearing (TC) procedure based the existing SUMIC method, and (Nature Cardiovascular Research, 2024, 3, 474–491; Cell, 2023, 186, 382–397.e24.). These researches have confirmed that the chemical reagents used in this method do not affect the inherent fluorescence signal of transgenic animals. With our improvements, we minimized the sample processing time as much as possible to avoid any potential adverse effects. The results in Figure 2, Figure 5, and Figure S1 indicated that after TC procedure, the tissue exhibit significant ZsGreen signals and certain tdTomato signals, which sufficiently support our conclusions.

(10) How did you address the issue of sample integrity and discontinuities in the z-axis caused by the stratification of slices in your reconstructions?

We greatly appreciate your comments. Currently, reconstruction techniques based on continuous sectioning cannot fully eliminate the discontinuities in the z-axis. Therefore, it is

for this reason that we need to compensate for this deficiency by imaging the whole tissue through TC procedure. These two 3D-reconstruction and imaging technologies complement each other to jointly address the spatial positioning and hierarchical relationships of PDGFR- α +, NFATc1+, and PDGFR- α +NFATc1+ cells from multiple perspectives. Additionally, this deficiency can be minimized by improving the technical skills, reducing section thickness, and to minimize tissue loss during sectioning, which is our future research endeavors.

(11) In Figure 2B, the schematic representation of the operational principle "Cre-loxp/Dre-loxp" does not correspond to the genotype "CreER/DreER". Please correct it.

Thanks for your important comments. We are sincerely sorry for the mistake. We have modified Figure 2B in the revised manuscript as below:

(12) Line 450, the specific distribution and differences of PDGFR- α +, NFATc1+, and PDGFR- α +NFATc1+ cells in pulp and periodontal tissues need to be further described and explained.

Thank you for your question. We have described this part on page 16 in the revised manuscript, "In PDL tissue, pulse data demonstrated widespread and abundant expression of PDGFR- α single-positive cells as well as NFATc1 single-positive cells, with no significant alteration in expression pattern or quantity after lineage tracing. Consequently, we conclude that in periodontal ligament and dental pulp tissues, PDGFR- α single-positive and NFATc1 single-positive cells primarily label intrinsic periodontal mesenchyme in PDL. Conversely, PDGFR- α +NFATc1+ cells exhibited a more confined localization in PDL. The tracing data clearly illustrated that PDGFR- α +NFATc1+ cells successfully gave rise to numerous progenies, which become predominant constituents within the periodontal ligament. In pulp tissue, the distribution of PDGFR- α single-positive cells was similar as that in PDL, primarily labeled odontoblast cell layer and there was not a significant increase in ZsGreen signal after tracing assay."

(13) In Figure S9, the sparse presence of NFATc1+ cells in pulp and periodontal tissue raises questions about the plasticity and differentiation potential of these cells. The author should include relevant discussions in this section.

Thanks for your suggestion. Considering the plasticity and differentiation potential of NFATc1+ cells, we conducted immunofluorescence staining and found that the PDGFR- α +NFATc1+ cell lineage in dental pulp and periodontal tissues represents a heterogeneous population. This population includes non-terminally differentiated mesenchymal stem cells (MSCs) as well as hematopoietic cells, indicating significant heterogeneity. We have also added this part of the discussion on page 17 of the manuscript.

Page 17 in the revised manuscript, "Cell ablation and immunofluorescence staining experiments further characterized the types and functions of PDGFR- α +/PDGFR- α +NFATc1+ populations. After ablating PDGFR- α + cells, we observed damage to the odontoblast layer and shrinkage of the pulp core in dental pulp tissue, indicating that PDGFR- α + cells contribute to the composition of dental pulp tissue, particularly the odontoblast layer (Figure. 9C, D). In the periodontal ligament, we noted a reduction and destruction of collagen fibers, suggesting a role for PDGFR- α + cells in periodontal tissue structure (Figure. 9E, F). Previous results confirmed the presence of double-positive cells in both dental pulp and periodontal tissues and provided insights into their hierarchical relationships in the periodontal ligament (Figure. 8). To further investigate the double-positive cell population, we developed an inducible dual-editing enzyme reporter system to label these cells with tdTomato signals. Using AlphaV as a marker for mesenchymal stem cells (MSCs) and CD45 for hematopoietic cells, we found that double-positive cells included components of both MSCs and

hematopoietic cells (Figure S22B, C), indicating a heterogeneous population. Further experiments are necessary to determine whether the predominant role in this co-positive MSC population is played by PDGFR- α ⁺ or NFATc1⁺ and to clarify the specific functions of these cells in the future.”

(14) Part 3, line 351, the authors were unable to confirm the hierarchical relationship between PDGFR- α ⁺ and NFATc1⁺ cells in the dental pulp region. Could this be due to limitations in experimental design or technical methods? Have you considered other factors that might explain these results?

Thank you for your question. We believe that the possible reason was that PDGFR- α ⁺ cells were a widely distributed constitutive component of dental pulp tissue, while NFATc1⁺ cells had a more limited expression range, resulting in a significant difference between the two. Therefore, we were unable to calculate the differences. In the future, we could further investigate the hierarchical relationship between the two by increasing the sample size or through in vitro experiments such as immunoprecipitation.

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